

A Data Science Analysis of Erdős Problem 30

The maximum size of a Sidon set in an interval

A computational and analytic investigation

John Akwei

Senior Data Scientist, ContextBase

johnakwei@gmail.com

with Anthropic Claude Fable 5.1 and Claude Opus 5.5

Stage 0 construction survey · the limit of the smoothing method, $\gamma^* = 2\sqrt{2}/3$ · the pair-level relaxation

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Executive summary

Erdős Problem 30 asks whether $h(N)$, the largest size of a Sidon set contained in $\{1, \dots, N\}$, satisfies $h(N) = N^{1/2} + O_\varepsilon(N^\varepsilon)$ for every $\varepsilon > 0$. The problem has been open since Erdős and Turán's 1941 paper and carries a \$1000 prize. The lower half of the conjecture reduces to prime gaps and is believed; the entire difficulty lies in the upper bound, where eighty-five years of work has moved the constant in front of $N^{1/4}$ and never the exponent.

This document collects the results of an investigation carried out in stages. Version 3 located the floor of the Hou–Zhao smoothing framework numerically at 0.942809, noted that it lies close to $2\sqrt{2}/3$, and proved a lower bound of 0.8991. Version 4 records the resolution of that question — the floor is $2\sqrt{2}/3$ exactly, proved from below for every certificate and attained from above — together with its extension to a relaxation that keeps what the smoothing argument discards. The principal findings are:

- **Stage 0 (construction survey, Part II).** Across the Singer, Bose and Ruzsa families, in every characteristic, over all prime powers and squares up to $N \approx 4 \times 10^6$, the excess $h(N) - \sqrt{N}$ produced by cutting a cyclic Sidon set at its largest gap grows logarithmically, matching a parameter-free extreme-value model. No family shows power-law behaviour. One genuine structural effect was found and explained: a Frobenius correlation in characteristic 2 that enlarges the constant but not the growth type.
- **The limit of the smoothing method (Parts III–V).** Every certificate of the vector-valued smoothing framework of Hou and Zhao — symmetric or two-sided, with any number of kernels, at any resolution — has $\gamma \geq 2\sqrt{2}/3 = 0.94280904\dots$. The proof uses one dual multiplier, a rescaled renewal measure of the uniform distribution, whose weight on the kernels' autocorrelation is exactly constant on its first window; the bound therefore depends on the kernels only through their energy a , and choosing the width of the multiplier gives $a \cdot b \geq 8/9$. For two-sided certificates the bound is sharp: the linear kernel $\rho(x) = 2x$, with boundary weights $e^t - 1$ at the end it faces and 1 at the other, attains $a \cdot b = 8/9$ in the continuum, and its discretizations approach it at rate $O(\log m/m^2)$. Whether symmetric certificates also reach $2\sqrt{2}/3$ is open; they are confined to $[2\sqrt{2}/3, 0.942810893]$.
- **A new certified constant (Part V).** The linear kernel at resolution $m = 512$ and boundary depth $L = 8$, checked in exact rational arithmetic on all 8194 covering inequalities, gives $h(N) \leq N^{1/2} + 0.9428096N^{1/4} + O(1)$. This is 6.8×10^{-4} below the constant 0.94349... of Hou and Zhao and 4.8×10^{-7} above the proved limit, so further optimization inside the framework cannot pay.
- **Keeping the variance term (Part VI).** The level-one moment relaxation of the Sidon condition keeps both quantities that the smoothing argument discards — the variance term in Cilleruelo's identity and the slack in the difference budget — yet has the same second-order constant: every smoothing certificate bounds it (proved), and explicit feasible points come within 1.03 of $\sqrt{N} + (2\sqrt{2}/3)N^{1/4}$ at $N = 10^4, 10^5, 10^6$ and 10^7 (computational). Three-point constraints lower the relaxation by an amount that shrinks with N , and in a cyclic model the four-point relaxation captures a vanishing share of the obstruction that separates it from genuine Sidon sets.

Dependence on the smoothing inequality. Every statement about $h(N)$ in this document rests on the smoothing inequality: Theorem 2.1 of Hou and Zhao (arXiv:2607.01169v3) for symmetric certificates, and its two-sided extension (Proposition 3.1 below) for the certificate of Part V. The two-sided version has now been proved in full, following Hou and Zhao step by step with the $O(1)$ term made explicit, and checked mechanically on explicit Sidon sets; neither proof has been refereed. The statements about the framework itself — the lower bound $2\sqrt{2}/3$, the extremal certificate, the verified certificates as feasible points — concern a finite optimization problem and its continuum limit, and do not depend on it.

Taken together, the evidence points the same way from both directions: the conjecture is probably true, the construction side will not refute it, and on the upper-bound side the smoothing/Cauchy–Schwarz lineage has now been followed to its exact end. An argument that goes below $2\sqrt{2}/3$ must use more than the one- and two-point statistics of the set — most plausibly its arithmetic structure.

Part I • The problem and its status

Statement

A Sidon set is a set of integers all of whose pairwise sums are distinct; equivalently all pairwise differences are distinct. Let $h(N)$ denote the maximum size of a Sidon set contained in $\{1, \dots, N\}$. Erdős Problem 30 asks whether, for every $\varepsilon > 0$, $h(N) = N^{1/2} + O_\varepsilon(N^\varepsilon)$.

Known bounds

- **Upper bound.** Erdős and Turán (1941) proved $h(N) \leq N^{1/2} + O(N^{1/4})$; their argument and Lindström’s give the constant 1. Since 2021 the constant has fallen in steps to 0.94349 (Hou–Zhao, 2026) and, in this work, to 0.9428096; Part V shows that $2\sqrt{2}/3$ is the limit of the method that produced the last two (Table 9).
- **Lower bound.** Singer’s construction gives a Sidon set of size $p+1$ in an interval of length p^2+p+1 for every prime p , so the lower-bound error is controlled entirely by gaps between primes near $\sqrt{N}$: unconditionally $N^{0.2625}$ (Baker–Harman–Pintz), $N^{1/4} \log N$ under RH, N^ε under Cramér-type hypotheses.
- The unconditional lower-bound exponent 0.2625 is therefore worse than the proved upper-bound exponent $1/4$ — an asymmetry that persists to this day.

Exact values

A Sidon subset of $\{1, \dots, N\}$ of size k is equivalent to a Golomb ruler with k marks and length at most $N-1$. Optimal Golomb ruler lengths are known exactly through 28 marks (OGR-28, length 585), so $h(N)$ is exactly determined for all $N \leq 586$. Exactness at the upper endpoint follows from strict monotonicity of the optimal lengths: a 29-mark ruler of length ≤ 585 would, on deleting an end mark, give a 28-mark ruler of length < 585 , contradicting OGR-28.

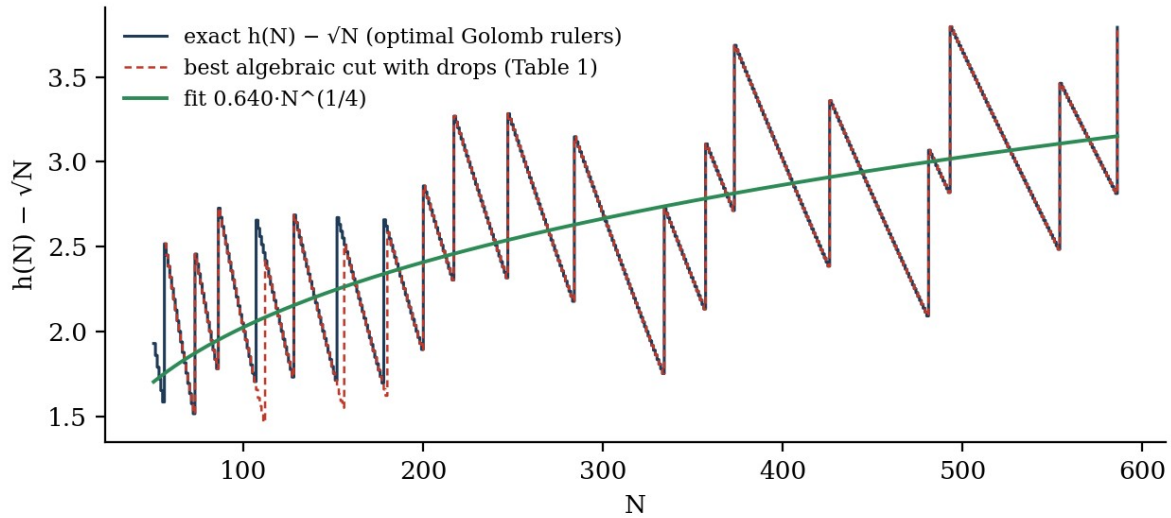


Figure 1. Exact $h(N) - \sqrt{N}$ from optimal Golomb rulers over $N = 50 \dots 586$, the best single-term fit $0.640N^{1/4}$, and the excess obtained from the best algebraic cut with drops of Table 1 (defined for $N \geq 56$). The two step functions coincide at all but 11 of the 531 values of N where both are defined and never differ by more than 1. The sawtooth is an artifact of $h(N)$ being a step function.

What the exact data supports

Fits over $N = 50 \dots 586$ (537 points): the model $\gamma N^{1/4}$ gives $\gamma = 0.640$ with RMSE 0.385; a constant gives $c = 2.64$ with RMSE 0.490; a free-exponent fit $A \cdot N^\alpha$ returns $\alpha = 0.20$ (0.201 by least squares on the values, 0.198 by least squares in log-log coordinates). The $N^{1/4}$ model beats the constant model and the free exponent lands nearer $1/4$ than 0, consistent with the gap still growing. But the residual floor is the step-function sawtooth, not model error, and over a range where $N^{1/4}$ varies by a factor of only 1.85 the candidate models are not separable.

The cleaner statistic is in k -space. Since $h(N) \leq \sqrt{N} + \gamma N^{1/4}$ is equivalent to $G(k) \geq k^2 - 2\gamma k^{3/2} + O(k)$, the deficit $D(k) = k^2 - G(k)$ gives one exact point per ruler with no sawtooth. Its log-log slope is 1.43 over $k = 10 \dots 28$, between the two models: the implied coefficient $D(k)/(2k^{3/2})$ drifts down (0.71, 0.64, 0.65, 0.58, 0.67 at $k = 10, 15, 20, 25, 28$) while $D(k)/(2k)$ drifts up (2.25, 2.47, 2.92, 2.90, 3.55).

Part II · Stage 0: the construction survey

Method

A cyclic Sidon set S in $\mathbb{Z}_n$ with a gap of length g can be rotated so the gap straddles the cut and read off as a Sidon set in an interval of length $n - g$. For Singer sets ($n = q^2 + q + 1$, $|S| = q + 1$) this gives $h(N) - \sqrt{N} \approx \frac{1}{2} + g/(2q)$ at $N = n - g$: the interval excess is the ratio of the largest gap to the mean gap. The question of whether that ratio grows logarithmically or as a power is therefore the whole conjecture, seen from the construction side.

A general engine was built over $\text{GF}(p^m)$ covering three families — Singer sets from the trace form in $\text{GF}(q^3)$, Bose sets from $\text{GF}(q^2)$, and Ruzsa sets in $\mathbb{Z}_{\{p(p-1)\}}$ — with every unit multiplier enumerated, and with the largest gap replaced by the full best-arc profile: the shortest interval retaining $k - d$ elements, for $d = 0 \dots 8$, since dropping a few elements may shorten more than cutting at one gap. Every generated set was verified Sidon before use.

Validation

The engine reproduces the known optimal-ruler lengths exactly at every k where an algebraic cut is optimal. The table below is the full comparison for $k = 10 \dots 28$.

k	OGR length	best algebraic cut	source
10	55	55	Singer $q=9$, drop 0
11	72	72	Singer $q=11$, drop 1
12	85	85	Singer $q=11$, drop 0
13	106	111 (+5)	Bose $q=13$, drop 0
14	127	127	Singer $q=13$, drop 0
15	151	155 (+4)	Singer $q=17$, drop 3
16	177	179 (+2)	Singer $q=17$, drop 2
17	199	199	Bose $q=17$, drop 0
18	216	216	Singer $q=17$, drop 0
19	246	246	Singer $q=19$, drop 1
20	283	283	Singer $q=19$, drop 0
21	333	333	Singer $q=23$, drop 3
22	356	356	Bose $q=23$, drop 1
23	372	372	Singer $q=23$, drop 1
24	425	425	Singer $q=23$, drop 0
25	480	480	Singer $q=25$, drop 1
26	492	492	Singer $q=25$, drop 0
27	553	553	Singer $q=27$, drop 1
28	585	585	Singer $q=27$, drop 0

Table 1. For every k from 17 to 28 the optimal Golomb ruler equals the best algebraic cut with drops; only $k = 13, 15, 16$ beat algebra, by 5, 4 and 2. The exact record above $N = 200$ is entirely the best-arc profile of algebraic sets.

Result: no power-law growth in any family

The extreme-value model treats each of the M inequivalent sets in a family as k uniform points on a circle of length n ; the largest gap over all of them is then the maximum of M Gumbel draws with scale q , predicting $\max \text{gap} \approx q[\ln k + \ln M]$. The table below compares observed to that parameter-free prediction.

family	n	EV ratio (mean $\pm$ sd)	slope of excess vs $\ln N$ (cut)	slope (best arc)	gain from drops	mean d
Singer, odd char	24	0.963 ± 0.108	0.843	1.019	0.70	2.4
Singer, char 2	6	1.129 ± 0.157	1.115	1.197	0.34	1.0
Bose, odd char	23	0.929 ± 0.081	0.765	0.957	0.87	2.2
Bose, char 2	6	1.115 ± 0.123	0.986	1.220	0.53	1.7
Ruzsa	108	1.029 ± 0.105	0.486	0.620	0.80	2.5

Table 2. Per-family summary, recomputed from `stage0_results.csv` over the instances with $q \geq 25$ (Singer, Bose) and $p \geq 53$ (Ruzsa); the four smallest Singer and three smallest Ruzsa instances in the file are excluded, which accounts for the counts $34 - 30$ and $111 - 108$. The standard deviation is the population value; slopes are least-squares slopes of the excess against $\ln N$. An EV ratio near 1 means the observed largest gap matches the random model with no fitted parameter. Ruzsa's lower slope is a search-size effect: multipliers give nothing beyond the $\varphi(p-1)/2$ primitive roots, so the family is much smaller than Singer's $\approx q^2/6$.

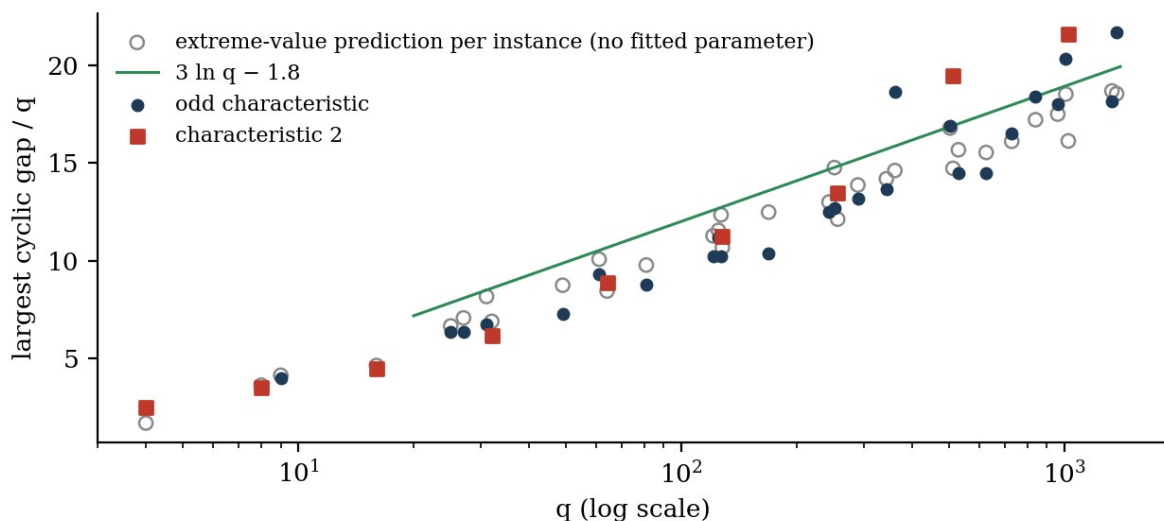


Figure 2. Largest cyclic gap in Singer sets, normalized by q , for the 34 Singer instances of `stage0_results.csv` ($4 \leq q \leq 1369$, every multiplier enumerated), against the extreme-value prediction for each instance, which has no fitted parameter, and its asymptotic form $3 \ln q - 1.8$. Characteristic 2 in red.

A separate gap-only run over 41 prime powers with $29 \leq q \leq 1409$, every multiplier enumerated, gives the following. Fitting $\max \text{gap} = Aq \ln^2 q + Bq \ln q + Cq$ returns $A = -0.23$: no super-logarithmic term. A naive power law returns exponent 1.26, but that is the local slope of $q \ln q$ over this range plus a finite-size deficit at small q . The Weil bound permits gaps up to $q^{3/2} \log q$; the actual sets use none of that room, with $\max \text{gap} / q^{3/2}$ falling monotonically from 1.2 to 0.5 across the range.

The one real structure: characteristic 2

Singer sets over $\text{GF}(2^e)$ have EV ratios rising 1.05, 1.11, 1.32, 1.34, 1.57 for $q = 2^7 \dots 2^{11}$, and a gap distribution with a tail 3–4 times heavier than Gumbel at the far quantiles, which odd characteristics lack. It persists at $q = 2^{11}$,

which has no proper subfields, so it is not Baer-subplane structure. Every extreme orbit has its gap centred on the origin.

The mechanism is Frobenius: the Singer condition $\text{Tr}(\theta^i) = 0$ satisfies $\text{Tr}(x^p) = \text{Tr}(x)^p$, so along the progression through the origin the zero events for j and pj coincide, cutting the number of independent constraints by a factor $(1 - 1/p)$. This predicts that the gap containing the origin has mean $2qp/(p-1)$ instead of $2q$. Measured:

q	p^e	mean (gap at origin) / q	prediction	$P(\text{gap}_0 > 6q)$
1009	1009^1	1.99	2.00	0.017
1024	2^{10}	3.95	4.00	0.198
625	5^4	2.50	2.50	0.043
961	31^2	2.09	2.07	0.021
729	3^6	1.49	1.50 (one-sided: $\text{Tr}(1)=0$)	0.017

Table 3. The Frobenius prediction verified to two decimals across characteristics. Non-origin gaps in characteristic 2 are statistically normal. The consequence is that in characteristic 2 the best cut has excess $\approx \ln N$ rather than $\approx \frac{3}{4} \ln N$ — still logarithmic.

Best-arc profile

Dropping d elements and taking the best arc gains 0.3–0.9 in excess on average, with the optimal d around 2–3; the largest single gain in stage0_results.csv is 2.6 (Ruzsa, $p = 599$). It never changes which family or which q wins by more than that. The largest excess found in the survey is 14.6, from the Singer set over $\text{GF}(2^{11})$ at $N \approx 4.1 \times 10^6$, in the extended Singer run; within stage0_results.csv, which reaches $N \approx 2.1 \times 10^6$, the largest is 12.36, from the Bose set over $\text{GF}(2^{10})$ with three elements dropped, at $N = 1,017,354$.

Verdict

Outcome A, cleanly. Every family follows the random model; the single place structure appeared, it was predictable to two decimals and enlarged a constant, not an exponent. The construction side will not refute the conjecture. The corresponding Cramér-type conjecture for this problem is that the largest gap over multipliers equals $q(3 \ln q + O(1))$ generically and $q(4 \ln q + O(1))$ in characteristic 2, so that $h(N) = \sqrt{N} + \Theta(\log N)$ along the best algebraic sequence — unbounded, hence refuting $h(N) = \sqrt{N} + O(1)$, but fully consistent with the N^ε conjecture.

Part III · Stage 2T(a): the smoothing framework, numerically

The Hou–Zhao certificate framework generalizes the upper-bound arguments of the Erdős–Turán lineage. This part sets up the framework and its two-sided extension and reports the numerical picture — resolution scans, the effect of asymmetry, the structure of optimal certificates, and the optimal dual multiplier — which is what led to the exact results of Parts IV and V. Version 3 located the infimum at 0.942809 and proposed $2\sqrt{2}/3$ as a candidate closed form; both are now theorems.

The certificate space

At resolution (m, L, K) — m bins per kernel, L boundary blocks, K kernels mixed with weights λ_k — a certificate consists of kernels $p^{(k)} \in \mathbb{R}^m$, symmetric, nonnegative, with $\sum_i p^{(k)}_i = 1$; boundary weights $w^{(k)} \in \mathbb{R}^{(Lm)}$, extended by $w^{(k)}_j = 1$ for $j \geq Lm$; and mixing weights $\lambda \geq 0$ with $\sum \lambda_k = 1$, subject to the covering inequalities

$$\sum_k \lambda_k \sum_i p^\wedge(k)_i w^\wedge(k)_{\{q+i\}} \geq 1 \quad \text{for } q = 0, \dots, Lm, \quad (3.1)$$

with

$$a = m \cdot \sum_k \lambda_k \sum_i (p^\wedge(k)_i)^2, \quad b = 1 + 2(\sum_k \lambda_k \sum_j (w^\wedge(k)_j)^2 / m - L), \quad \gamma = \sqrt{ab}. \quad (3.2)$$

Theorem 2.1 of Hou and Zhao. If (3.1) holds and $b > 0$, then $h(N) \leq N^{1/2} + \sqrt{ab} N^{1/4} + O(1)$. Version 3 cited this as Lemma 2.1; the current version of arXiv:2607.01169 (v3, 4 September 2026) states it as Theorem 2.1 and certifies $\gamma_0 = 0.9434925907\dots$ with eight kernels at $m = 32$, $L = 4$. The uniform kernel with all boundary weights 1 gives $a = b = 1$ and recovers Erdős–Turán–Lindström. The hypothesis $b > 0$ is automatic: Hou and Zhao show $b \geq 1/m$, and Part IV gives $b \geq 8/(9a)$.

The optimum is non-increasing in L and K — extra boundary weights can be set to 1 and extra kernels given weight 0 — and under refinement $m \rightarrow 2m$, since splitting each bin in two preserves feasibility exactly. Write γ^*_{sym} for the infimum of $\sqrt{ab}$ over all certificates at all resolutions.

Two-sided certificates

The proof of Theorem 2.1 uses the symmetry of the kernels at one point only: in the right-boundary case of the covering claim, where $K_s = K_{[H-1-s]}$ reduces that case to the left one. Dropping it gives a two-sided certificate: kernels $p^\wedge(k)$ that are nonnegative and normalized but need not be symmetric, and for each k two boundary vectors $u^\wedge(k), v^\wedge(k) \in \mathbb{R}^\wedge(Lm)$, extended by 1, subject to

$$\begin{aligned} \sum_k \lambda_k \sum_i p^\wedge(k)_i u^\wedge(k)_{\{q+i\}} &\geq 1 \quad (q = 0, \dots, Lm), \\ \sum_k \lambda_k \sum_i p^\wedge(k)_{\{m-1-i\}} v^\wedge(k)_{\{q+i\}} &\geq 1 \quad (q = 0, \dots, Lm), \end{aligned} \quad (3.3)$$

with a as before and $b = 1 + (1/m) \sum_k \lambda_k \sum_{j < Lm} [(u^\wedge(k)_j)^2 + (v^\wedge(k)_j)^2 - 2]$. A symmetric certificate with $u = v = w$ is a two-sided certificate with the same a and b .

Proposition 3.1 (smoothing inequality, two-sided). If a two-sided certificate satisfies (3.3) and $b > 0$, then $h(N) \leq N^{1/2} + \sqrt{ab} N^{1/4} + O(1)$. Explicitly, with $\gamma = \sqrt{ab}$ and $N_1 = N + bm$, for $N \geq N_0(a, b, m, L)$,

$$h(N) \leq \sqrt{N_1} + \gamma N_1^{1/4} + \gamma^2/2 + \gamma^3/(8N_1^{1/4}) = \sqrt{N} + \gamma N^{1/4} + ab/2 + o(1). \quad (3.4)$$

The proof follows Hou and Zhao step by step, using $v^\wedge(r)$ and the reversed kernel on the right boundary, and replaces their final choice of scale, which calls on the Erdős–Turán bound, by the self-contained inequality $k \leq \sqrt{N_1} + \gamma\sqrt{k}$. It was implemented literally at integer positions and checked on six certificates, five explicit Sidon sets (Ruzsa sets with 30, 46 and 60 elements, an Erdős–Turán set and a greedy random set) and $h = 1, 2, 3$: every check passed, and swapping u and v makes the covering fail, its minimum dropping to 0.44 for the linear kernel. Write γ^* for the infimum over two-sided certificates, so that $\gamma^* \leq \gamma^*_{\text{sym}}$.

The continuum relaxation

Reading kernels and boundary vectors as step functions on cells of width $1/m$ — $\rho_k = m p^\wedge(k)_i$ on $[i/m, (i+1)/m)$, and $\omega_k = u^\wedge(k)_j$, $\omega'_k = v^\wedge(k)_j$ on $[j/m, (j+1)/m)$ — turns every discrete certificate into a continuum certificate: probability densities ρ_k on $[0, 1]$ and profiles ω_k, ω'_k on $[0, \infty)$, equal to 1 outside a compact set, such that

$$\sum_k \lambda_k \int_0^1 \rho_k(x) \omega_k(u+x) dx \geq 1 \quad \text{and} \quad \sum_k \lambda_k \int_0^1 \bar{\rho}_k(x) \omega'_k(u+x) dx \geq 1 \quad \text{for all } u \geq 0, \quad (3.5)$$

where $\bar{\rho}(x) = \rho(1-x)$, with $a = \sum_k \lambda_k \int_0^1 \rho_k^2$ and $b = 1 + \sum_k \lambda_k \int_0^\infty (\omega_k^2 + \omega'_k{}^2 - 2)$. At a fractional shift $u = (q+s)/m$ the covering integral equals $(1-s)C_q + sC_{\{q+1\}}$, a convex combination of two adjacent discrete constraints, and a and b take identical values in both settings. Consequently every lower bound for the continuum problem applies verbatim to every discrete certificate at every resolution, with no limiting argument; this is how the bound of Part IV reaches the discrete framework.

Method

- **Outer:** L-BFGS over the kernels (softmax-parametrized; in the symmetric case reduced to $m/2$ free parameters per kernel) and the mixing weights.
- **Inner:** the exact quadratic program for the boundary weights, solved through its nonnegative dual, which is banded with bandwidth m , by a primal–dual active-set iteration warm-started from the previous outer step; for two-sided certificates it splits into one program per end. The gradient of the outer objective is read off the duals by the envelope theorem.
- **Globalization:** multi-start with 30+ random seeds at $m = 32$ plus basin hopping; higher resolutions are initialized by upsampling the converged lower-resolution certificate. Splitting every bin in two preserves feasibility and γ , but it also produces an exact stationary point of the finer problem, so the upsampled certificate must be perturbed before re-optimization. The $m = 512$ symmetric value of Version 3 (0.94281112) was under-converged for this reason; its re-optimized successor gives 0.9428108915.

The analytic gradient agrees with central finite differences to 5.5×10^{-6} , and an independent implementation of the two-sided optimizer agrees to a relative 3×10^{-8} . At Hou–Zhao’s grid ($m = 32$, $L = 4$) with one symmetric kernel the optimizer returns 0.946147, the value 0.9461473014 listed in their Table 1. Every reported certificate value is verified independently of the optimizer: the certificate is rationalized, the boundary program re-solved for the rational kernels, the boundary weights rounded and repaired by a uniform increment (Hou–Zhao, Lemma 3.2), all covering inequalities checked over $\mathbb{Q}$, and a , b and γ^2 evaluated exactly.

Resolution scans

m	symmetric, $K = 2$, $L = 4$	two-sided, $K = 2$, $L = 4$	linear kernel $2 \times$, $L = 8$
32	0.94324229	0.9429206688	0.9429240903
64	0.94292072	0.9428374024	0.9428378328
128	0.94283748	0.9428162103	0.9428162572
256	0.94281628	0.9428108643	0.9428108625
512	0.9428108915	—	0.9428095137
Richardson (last two)	0.94280910	0.94280908	0.94280906

Table 4. Verified values for the optimized certificates — the symmetric $m = 512$ entry is the re-optimized certificate; the original gave 0.94281112 — and exact quadratic-program values for the linear kernel, whose $m = 256$ and 512 certificates were also verified in exact arithmetic (Table A1). The last row is two-point Richardson extrapolation in $1/m^2$ from the last two resolutions. For comparison, $2\sqrt{2}/3 = 0.94280904$.

All three scans converge in $1/m^2$: successive drops fall by factors 3.86, 3.93 and 3.93 (symmetric), 3.93 and 3.96 (two-sided), and within 3×10^{-3} of 4 at every step for the linear kernel. The two $L = 4$ scans extrapolate to 0.9428091, about 4×10^{-8} above $2\sqrt{2}/3$. That residual is the cost of the four-width boundary box: the extremal certificate of Part V needs a boundary layer about six kernel widths deep, and at $L = 4$ the linear kernel loses about 10^{-4} in b , which optimized mixtures recover almost entirely. At $L = 8$ the linear kernel extrapolates to 0.94280906 and follows $a \cdot b = 8/9 + 2/(9m^2) - 1/(9m^4)$ to within 3×10^{-8} in b (Table 8). Version 3 proposed an $m = 1024$ certificate to separate $2\sqrt{2}/3$ from the Richardson value 0.9428094; that test is superseded by the proofs of Parts IV and V.

Saturation in boundary depth and number of kernels

For boundary depth, at $m = 32$ and $K = 2$, $L = 3$ gives 0.943245 while $L = 4, 5$ and 6 all give 0.943242 — the boundary layer of the symmetric optimum is under three kernel widths deep. For the number of kernels, $K = 1$

gives 0.946147, $K = 2$ gives 0.943242, and $K = 3$ gives 0.943242 again, identical to five decimals at both $m = 32$ and $m = 64$, with the optimizer driving one weight to about 0.04. Within the symmetric parametrization the capacity saturates at two kernels, which buy 2.9×10^{-3} over one. The saturation in L is a property of these optima; it fails for the extremal two-sided certificate of Part V, whose boundary profile needs about six kernel widths.

Asymmetry

$L = 4$	$m = 32$	64	128	256	limit
symmetric, $K = 1$	0.946147	0.945669	0.945610	—	≈ 0.94559
two-sided, $K = 1$	0.942944	0.942855	0.942832	—	≈ 0.942825
symmetric, $K = 2$	0.943242	0.942921	0.942837	0.942816	0.9428091
two-sided, $K = 2$	0.942921	0.942837	0.942816	0.942811	0.9428091

Table 5. Symmetric against two-sided certificates at $L = 4$. The $K = 2$ rows and the $m = 32$ entries are verified values; the remaining $K = 1$ entries are optimizer values. Limits are Richardson extrapolations from the last two resolutions shown; for the symmetric single kernel the convergence is irregular and the limit is rough.

For a single kernel, asymmetry lowers the limit by almost 3×10^{-3} , to within 2×10^{-5} of the two-kernel floor: the single-kernel value 0.946147 is a property of the symmetric parametrization rather than of one kernel as such. For two kernels, asymmetry does not move the $L = 4$ limit; a two-sided certificate at resolution m does as well as a symmetric one at $2m$, slightly better by $5\text{--}8 \times 10^{-8}$. Part V explains the pattern: the extremal kernel is asymmetric, symmetric certificates can approach it only through mixtures, and with a deeper boundary layer a single asymmetric kernel reaches the limit on its own. Decoupling the kernel grid from the boundary grid cannot move the infimum at all, since such a certificate is an ordinary certificate on the common refinement of the two grids, and the continuum relaxation has no grid.

Structure of the optimal certificates

The symmetric optimum is a two-kernel mixture with $\lambda \approx (0.34, 0.66) \text{ --- } (0.3396, 0.6604)$ at $m = 512$ in the original scan, $(0.3383, 0.6617)$ after re-optimization. The kernel carrying weight 0.34 is narrow and spiked, with peak density 2.69 and a peak-to-edge ratio of 17.3; the kernel carrying weight 0.66 is broad and nearly flat, with peak 1.87 and ratio 9.0. Both are symmetric and unimodal. Their boundary weights dip to roughly 0.08–0.10 at the outermost cell, overshoot above 1 near half a kernel width (reaching 1.37), and oscillate back toward 1 by about three kernel widths. The overshoot is what allows the covering inequality to hold while the outermost cells are heavily discounted. That the profile is confined to a few kernel widths is the dual-side statement that the method sees only a boundary layer of depth $O(N^{(3/4)})$ — the same scale as Cilleruelo’s gap theorem, reached independently.

The two-sided optimum at $L = 4$ is organized differently: $\lambda \approx (0.15, 0.85)$ at every resolution from $m = 32$ to 256; the lighter kernel is strongly asymmetric, at total variation distance 0.67 from its reversal, while the heavier one is close to symmetric, at 0.19.

The optimum has flat directions. Along the curve of optimal certificates at prescribed a , the product $a \cdot b$ changes by less than 10^{-4} as a runs from 1.32 to 2.0, and at $m = 256$ the constants a and b of the two-sided optimum drift by about 10^{-3} over the last few hundred iterations while γ changes by 10^{-9} . Only γ is well determined; the individual a and b in Table A1 should not be read as converged.

The optimal dual multiplier

For fixed kernels the boundary problem is a quadratic program whose dual solution is a nonnegative measure μ on the shifts $u \geq 0$ — the multiplier of Part IV, now optimal rather than chosen. At the optimal certificates every covering constraint is active, and the optimality conditions read

$$(\mu \star R)(u) = \int_0^\infty R(u - v) d\mu(v) = 4 \quad \text{for all } u \geq 0, \quad (3.6)$$

where $R = \sum_k \lambda_k \rho_k \star \tilde{\rho}_k$ is the mixed autocorrelation of the kernels, $\tilde{\rho}(x) = \rho(-x)$, normalized so that the flat multiplier is 4Leb . This is a Wiener–Hopf equation of the first kind; for two-sided certificates each end has its own multiplier and both satisfy (3.6) with the same R . The one-atom multiplier of Version 3 is its solution at $u = 0$ when the continuous part is flat: $4^{1/2} + \beta R(0) = 4$ gives $\beta = 2/a$. Numerically, the optimal multiplier at the two-sided certificates converges, at rate $1/m$, to

$$\mu_1 = 2 \sum_{n \geq 0} U^\wedge(\cdot + n), \quad (3.7)$$

twice the renewal measure of the uniform distribution U on $[0,1]$: an atom of mass 2 at the origin, density $2e^\wedge u$ on $[0,1]$, density $2(e^\wedge u - u e^\wedge(u-1))$ on $[1,2]$, and so on, settling to the flat value 4 with oscillations that decay like $e^\wedge(-2.09u)$ (Figure 4a). At resolution m the atom has mass $2 - 2/m$ and the density just after it is $2(1 + 1/m)$; the mean deviation from (3.7) on $[0,1]$ is 0.010, 0.005 and 0.003 at $m = 64, 128$ and 256 . The renewal identity $\mu_1 = 2\delta_0 + U \star \mu_1$ explains the flat value: the bulk density is the atom, 2, divided by the mean $1/2$ of U . Feeding μ_1 back into (3.6) determines R on $[0,1]$, and hence the kernel: the linear density $2x$ (Part V).

multiplier	bound on $a \cdot b$	share of the gap
flat, 4Leb	0.3748	0%
flat + atom at 0 (Version 3)	0.8580	94.00%
+ linear ramp on $[0,1]$	0.88809	99.84%
+ linear piece on $[1,2]$	0.88876	99.97%
exact (optimal multiplier)	0.88889	100%

Table 6. Truncations of (3.7) evaluated at the optimal two-sided certificate ($m = 256, L = 4$), whose exact value is $ab = 0.888892$, each family fitted by maximizing the dual value over its parameters. These are ceilings for what each family could prove on that certificate; Part IV shows that the full renewal multiplier, rescaled with the certificate, proves $ab \geq 8/9$ for every certificate.

Before the proof was known, a three-parameter family $\mu = 4\text{Leb} + \beta\delta_0 + \Theta(2r(u/w) - 4)$, fitted per certificate under $\mu \geq 0$, was searched for its worst kernel over single kernels and two-kernel mixtures at $m = 32, 64, 128$ and 256 . The worst certified $a \cdot b$ fell to 0.8890087, 0.8889188, 0.88889636 and 0.88889075, with drop ratios 4.006 and 4.000, extrapolating to 0.888888880. At the minimizers — mildly U-shaped kernels, not the linear one — the family selected $\beta = 2w$, $\Theta = 1$ and $aw = 4/3$. That is exactly the multiplier of the proof in Part IV, which explains why the fitted family never certified less than $8/9$.

Caveats

- The value of γ^* is no longer a numerical claim: the lower bound (Part IV) and the matching two-sided certificates (Part V) are proofs, and every certificate value in Table 4 is exact-rational verified or an exact quadratic-program value. What remains numerical is the global optimality of the optimized symmetric certificates, and hence anything said about the symmetric floor beyond the rigorous bracket $[2\sqrt{2}/3, 0.942810893]$.
- The optimized certificates have flat directions; only γ is converged.
- Only the implication to $h(N)$ depends on the smoothing inequality; nothing in Parts III–V about the framework itself does.

Part IV · The lower bound $\gamma \geq 2\sqrt{2}/3$

This part proves, with no numerics in the argument, that the infimum located in Part III is at least $2\sqrt{2}/3$. Version 3 proved the weaker bound $\sqrt{((\pi^2+16)/32)} = 0.8991$ with a one-atom multiplier; that argument survives here as a stepping stone, and the gap it left is closed by replacing the atom with the full renewal measure (3.7), rescaled with the certificate.

Theorem 4.1 (lower bound). Every certificate — discrete or continuum, symmetric or two-sided, with any number of kernels — satisfies $a \cdot b \geq 8/9$, that is $\gamma = \sqrt{ab} \geq 2\sqrt{2}/3 = 0.94280904\dots$. Consequently $\gamma^*_{\text{sym}} \geq \gamma^* \geq 2\sqrt{2}/3$, and no certificate of the framework proves $h(N) \leq N^{1/2} + \gamma N^{1/4} + O(1)$ with $\gamma < 2\sqrt{2}/3$.

By the continuum reduction of Part III it suffices to prove the theorem for two-sided continuum certificates.

A general dual bound

For a continuum certificate write $R = \sum_k \lambda_k \rho_k * \tilde{\rho}_k$ for the mixed autocorrelation. R is even, nonnegative, supported in $[-1, 1]$, with $\int R = 1$ and $R(0) = a$. Put

$$G = \int |s| R(s) ds, \quad T(v) = \int_{-\infty}^{\infty} v^\infty R(s) ds \quad (v \geq 0). \quad (4.1)$$

A multiplier is a nonnegative measure on $[0, \infty)$ of the form $\mu = 4\text{Leb} + \nu$, with ν a finite signed measure.

Lemma 4.2 (dual bound). For every continuum certificate and every multiplier $\mu = 4\text{Leb} + \nu$,

$$b \geq G + \int T d\nu - \frac{1}{8} \iint R(u - v) \nu(du) \nu(dv). \quad (4.2)$$

Equality holds if both covering conditions are equalities on the support of μ and $\omega_k = \frac{1}{4}(\mu * \rho_k)$, $\omega'_k = \frac{1}{4}(\mu * \tilde{\rho}_k)$ for every k .

Proof. Write $e_k = \omega_k - 1$, so that the first end contributes $\sum_k \lambda_k \int (2e_k + e_k^2)$ to $b - 1$ and its covering condition reads $\sum_k \lambda_k \int \rho_k(x) e_k(u+x) dx \geq 0$. For fixed kernels the minimization of b is a convex quadratic program, so integrating this constraint against $\frac{1}{2}\mu \geq 0$ and subtracting gives a valid lower bound by weak duality. By Fubini the result is a pointwise problem in $e_k(t)$, minimized at $e_k = \frac{1}{4}(\mu * \rho_k) - 1$ with value $-\sum_k \lambda_k \int_{-\infty}^{\infty} (\frac{1}{4}(\mu * \rho_k) - 1)^2$. Since $4\text{Leb} * \rho_k = 4C_k$, with C_k the distribution function of ρ_k , the integrand is $(\frac{1}{4}(\nu * \rho_k) - (1 - C_k))^2$. Expanding: $\int (\nu * \rho_k)^2 = \iint r_k(u - v) \nu(du) \nu(dv)$ with $r_k = \rho_k * \tilde{\rho}_k$; $\int (\nu * \rho_k)(1 - C_k) = \int P(Y_k - X_k > u) \nu(du)$ for X_k, Y_k independent with density ρ_k ; and $\int (1 - C_k)^2$ is a constant. At the second end ρ_k is replaced by $\tilde{\rho}_k$, which changes neither r_k nor $P(Y_k - X_k > u)$, because X_k and Y_k are exchangeable, and turns the constant into $\int C_k^2$. The two constants sum to $1 - 2\int C_k(1 - C_k) = 1 - E|X_k - Y_k|$. Summing over the ends and the kernels gives (4.2). ■

Two consequences are worth recording. The right-hand side of (4.2) depends on the kernels only through R , so asymmetric kernels matter only through the autocorrelations they produce; in the discrete boundary program, where strong duality holds, the same is true of the optimal b itself, and reversing any kernel of a two-sided certificate changes b by less than 10^{-13} in computation. And the one-atom multiplier recovers Version 3's bound, as follows.

The one-atom bound and the cosine-arch inequality

With $\nu = \beta\delta_0$ one has $\int T d\nu = \beta T(0) = \beta/2$ — this is the kernel-independent cross term of Version 3, $\int_{-\infty}^{\infty} (1 - C(t)) \rho(t) dt = \int_0^1 (1 - C) dC = \frac{1}{2}$ — and $\frac{1}{8} \iint R \nu \nu = \beta^2 a/8$. So (4.2) gives $b \geq G + \beta/2 - \beta^2 a/8$, which is maximized at $\beta = 2/a$:

$$b \geq G + 1/(2a), \quad a \cdot b \geq a \cdot G + \frac{1}{2}. \quad (4.3)$$

The mixture collapses by Cauchy–Schwarz in the weights: with $F_k = \|\rho_k\|_{2^2}$ and $G_k = E|X_k - Y_k|$, $aG = (\sum_k \lambda_k F_k)(\sum_k \lambda_k G_k) \geq (\sum_k \lambda_k \sqrt{F_k G_k})^2 \geq \min_k F_k G_k$. The single-kernel product obeys a sharp inequality.

Lemma 4.3 (cosine-arch inequality). For every probability density ρ on $\mathbb{R}$ with $\rho \in L^2$, $Q(\rho) := \|\rho\|_{2^2} \cdot E|X - Y| \geq \pi^2/32 = 0.3084251\dots$, with equality precisely for the cosine arches $\rho_W(x) = (\pi/4W) \cos(\pi x/2W)$ on $[-W, W]$, $W > 0$, and their translates.

Proof. Let $x(u)$ be the quantile function of ρ and $g = x'$, so that $\rho(x(u))g(u) = 1$ for almost every u . The substitution $u = F(x)$ gives $\|\rho\|_{2^2}^2 = \int_0^1 du/g(u)$ and $E|X - Y| = 2 \int_0^1 u(1-u) dx(u) \geq 2 \int_0^1 u(1-u) g(u) du$, the singular part of dx — supported on gaps in the support of ρ — contributing nonnegatively. By Cauchy–Schwarz, $(\int_0^1 du/g)(\int_0^1 u(1-u) g du) \geq (\int_0^1 \sqrt{u(1-u)} du)^2 = (\pi/8)^2$, the integral being the area of a half-disc of radius $1/2$; hence $Q(\rho) \geq 2(\pi/8)^2 = \pi^2/32$. Equality forces the singular part to vanish and $g(u) = C/\sqrt{u(1-u)}$, so $x(u) = c + C \arcsin(2u - 1)$: the cosine arch with $W = \pi C/2$, translated to c . Conversely $\|\rho_W\|_{2^2}^2 = \pi^2/(16W)$ and $E|X - Y| = W/2$. ■

The cosine arch is also the stationary point of Q , but after normalizing $E|X - Y| = 1$ the constraint set $\{E|X - Y| \leq 1\}$ is a sublevel set of a concave functional and is not convex, so stationarity alone does not prove minimality; the quantile argument avoids the issue. The inequality is elementary and no claim of novelty is made for it.

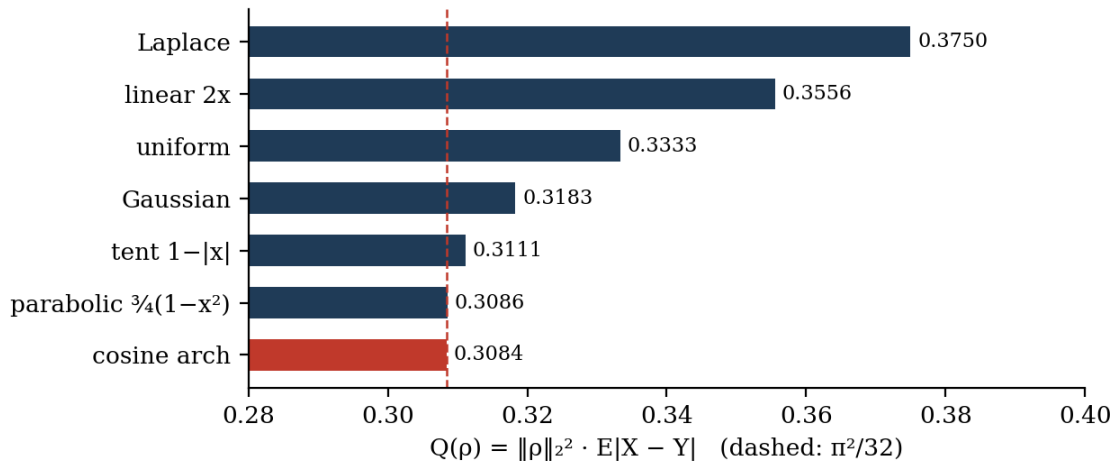


Figure 5. The scale-invariant functional Q of Lemma 4.3 on standard densities. The cosine arch attains $\pi^2/32$; the parabolic density comes within 1.5×10^{-4} of it, and the others lie higher.

Combining, every certificate satisfies $a \cdot b \geq \pi^2/32 + 1/2 = (\pi^2 + 16)/32$, that is $\gamma \geq 0.89912465\dots$ — the theorem of Version 3, which by Lemma 4.2 now holds for two-sided certificates as well. Each inequality in the chain is sharp on its own (the cosine arch attains Lemma 4.3, and a single kernel makes the mixture step an equality), so 0.8991 is the best that the one-atom family can give, and the remaining gap of 0.0437 in γ lies entirely in the choice of multiplier. Table 7 shows how each multiplier family performs on reference kernels.

kernel	a = F	F·G (flat)	F·G + ½ (one atom)	renewal $\mu_w, w = 4/(3a)$	F·B (exact)
uniform on [0,1]	1	$1/3 \approx 0.333$	$5/6 \approx 0.833$	$8/9 \approx 0.889$	1.000
cosine arch	$\pi^2/8 \approx 1.234$	$\pi^2/32 \approx 0.308$	$(\pi^2+16)/32 \approx 0.808$	$8/9 \approx 0.889$	0.982
linear 2x	$4/3$	$16/45 \approx 0.356$	$77/90 \approx 0.856$	$8/9 \approx 0.889$	$8/9 \approx 0.889$
V3 certificate, kernel 0 (m = 512)	—	≈ 0.369	0.869	≥ 0.889	0.922
V3 certificate, kernel 1 (m = 512)	—	≈ 0.333	0.833	≥ 0.889	0.959

Table 7. Lower bounds on ab for single kernels from successively richer multipliers, against the exact inner value FB , where B is the minimum of b over boundary profiles for that kernel. By (4.3) the one-atom column is exactly the flat column plus $\frac{1}{2}$. The renewal column is exactly $8/9$ whenever $a \leq 4/3$ (see the remark after Theorem 4.7): the renewal multiplier cannot distinguish kernels of equal energy, and it is tight only for the linear kernel. All entries are closed forms except the cosine arch's FB (0.9820 on grids of 64, 128 and 256 bins) and the certificate rows, whose FG values are obtained from the directly evaluated one-atom values through (4.3). For the uniform kernel $B = 1$ exactly, since summing the covering condition over integer shifts gives $\int_0^\infty e \geq 0$ at each end.

The renewal density of the uniform distribution

Let U be the uniform density on $[0,1)$ and $M = \sum_{n \geq 0} U^\wedge(\star n)$ its renewal measure: an atom of mass 1 at the origin plus a density r . Put $g = r - 2$ and $G_1(t) = \int_0^t g$.

Lemma 4.4. (a) $r(x) = e^\wedge x$ on $[0,1)$, $r(x) = e^\wedge x - x e^\wedge(x-1)$ on $[1,2)$, and $r(x) = \int_{-1}^x r(t) dt$ for $x \geq 1$. (b) g decays exponentially: $|g(x)| \leq C_0 e^\wedge(-\kappa x)$ for some $C_0, \kappa > 0$; in particular $g \in L^1 \cap L^2$. (c) $G_1(t) \geq -1$ for every $t \geq 0$, and $\int_0^\infty g = -1/3$.

Proof. (a) is the renewal equation $r = U + U \star r$, solved on $[0,1)$ and $[1,2)$ by the method of steps. For (b), the Laplace transform of g is $s\hat{q}(s)/(s - 1 + e^\wedge(-s))$ with $q(x) = (2x - 1) \cdot \mathbb{1}_{[0,1)}(x)$. It is analytic in $\text{Re } s > -2.0888$, because the non-zero roots of $s = 1 - e^\wedge(-s)$ all have real part at most -2.0888 and the double zero of the denominator at $s = 0$ is cancelled by the numerator, since $\hat{q}(0) = \int q = 0$; exponential decay then follows from standard renewal theory for distributions with a density and finite exponential moments (Stone 1965; Feller, Ch. XI). For (c), $\int_0^\wedge t r$ is the renewal function $H(t)$, and Wald's identity gives $H(t) + 1 \geq t/E[U] = 2t$. Finally $\int_0^\infty g$ is the Laplace transform at the origin, where $\hat{q}(s) = -s/6 + O(s^2)$ and $s - 1 + e^\wedge(-s) = s^2/2 + O(s^3)$. ■

Lemma 4.5 (autocorrelation of the renewal density). Let $A(t) = \int_0^\infty g(x) g(x+t) dx$. Then $A(t) = e^\wedge t - 2t - 2/3$ for $0 \leq t \leq 1$. In particular $\int_0^\infty (r(x) - 2)^2 dx = 1/3$.

Proof. By Lemma 4.4(a), $g = U \star g + q$, so in the sense of distributions on $(0, \infty)$, $g' = g - g(\cdot - 1) \mathbb{1}_{\{x \geq 1\}} + 2 \cdot \mathbb{1}_{(0,1)} - \delta_1$, where δ_1 records the jump of g by -1 at $x = 1$. For $0 < t < 1$, differentiating A under the integral and splitting at $x = 1 - t$, where $g(x+t)$ jumps, gives the functional equation

$$A'(t) = A(t) - A(1-t) + e^\wedge(1-t) + 4t - 4. \quad (4.4)$$

Put $D(t) = A(t) - A(1-t)$. Applying (4.4) at t and at $1-t$ gives $D'(t) = e^\wedge t + e^\wedge(1-t) - 4$ with $D(1/2) = 0$, so $D(t) = e^\wedge t - e^\wedge(1-t) - 4t + 2$; then (4.4) reads $A'(t) = e^\wedge t - 2$, whence $A(t) = e^\wedge t - 2t + C$ on $[0,1]$. For $t > 1$ the same differentiation gives the delay equation $A'(t) = A(t) - A(t-1)$, for whose solutions $\Phi(t) = A(t) - \int_{-1}^t A$ and $\Psi(t) = \int_{-1}^t (s-t+1)A(s) ds$ satisfy $\Phi' = 0$ and $\Psi' = \Phi$ on $(1, \infty)$. Since $g \in L^2$, $A(t) \rightarrow 0$, so $\Phi = 0$ and then $\Psi \equiv 0$. Now $\Psi(1) = \int_0^1 s A(s) ds = 1/3 + C/2$, so $C = -2/3$, and $\int g^2 = A(0) = 1/3$. ■

Lemma 4.6 (weight identity). Let $\nu_1 = 2M - 4\text{Leb} = 2\delta_0 + 2g$, and for $s > 0$ put

$$h(s) = 2s + \nu_1([0, s]) - \frac{1}{4} \int \nu_1(du) \nu_1(d(u+s)). \quad (4.5)$$

Then $h(s) = 8/3$ for $0 < s < 1$, and $h(s) \geq 8/3$ for $s \geq 1$.

Proof. Here $\nu_1([0, s]) = 2 + 2G_1(s)$, and for $s > 0$ the double integral is $4g(s) + 4A(s)$, an atom–density term plus a density–density term, so

$$h(s) = 2s + 2 + 2G_1(s) - g(s) - A(s). \quad (4.6)$$

For $0 < s < 1$, Lemmas 4.4(a) and 4.5 give $g(s) = e^s - 2$, $G_1(s) = e^s - 1 - 2s$ and $A(s) = e^s - 2s - 2/3$; substituting, every term cancels except $8/3$. For $s \geq 1$, Lemma 4.4(a) gives $g(s) = G_1(s) - G_1(s - 1)$, so $h(s) = 2s + 2 + G_1(s) + G_1(s - 1) - A(s)$, and $A(s) \leq A(0) = 1/3$ by Cauchy–Schwarz. If $s \geq 3/2$, Lemma 4.4(c) gives $h(s) \geq 2s - 1/3 \geq 8/3$. If $1 \leq s < 3/2$, then $g \geq e - 3$ on $[1, 3/2]$, since r increases there from $r(1+) = e - 1$, so $G_1(s) \geq (3/2)(e - 3)$; and G_1 is decreasing on $[0, \ln 2] \supset [0, 1/2]$, so $G_1(s - 1) \geq e^{1/2} - 2$. Together, $h(s) \geq 4 + (3/2)(e - 3) + e^{1/2} - 2 - 1/3 = 2.8928... > 8/3$. ■

The weight jumps at $s = 1$ to $h(1+) = 11/3$ and increases thereafter (Figure 4c).

Proof of the lower bound

For $w > 0$ let μ_w be the renewal multiplier rescaled to width w ,

$$\mu_w = 2w \cdot \sum_{n \geq 0} U_w^{(*n)}, \quad \mu_w(du) = 2w\delta_0 + 2r(u/w) du, \quad (4.7)$$

with U_w the uniform distribution on $[0, w]$: an atom of mass $2w$, density $2e^{(u/w)}$ on $[0, w]$, settling to 4. By Lemma 4.4(b), $\nu_w = \mu_w - 4\text{Leb}$ is a finite signed measure.

Theorem 4.7. Every continuum certificate satisfies $b \geq 4w/3 - (w^2/2)a$ for every $w > 0$. Taking $w = 4/(3a)$ gives $b \geq 8/(9a)$, that is $a \cdot b \geq 8/9$.

Proof. Apply Lemma 4.2 with $\mu = \mu_w$. Since R is even, $G = \int_0^\infty 2s R(s) ds$, $\int T d\nu_w = \int_0^\infty R(s) \nu_w([0, s]) ds$ by Fubini, and $\iint R(u - v) \nu_w(du) \nu_w(dv) = (2w)^2 R(0) + 2 \int_0^\infty R(s) \int \nu_w(du) \nu_w(d(u+s)) ds$. The right-hand side of (4.2) is therefore

$$\int_0^\infty R(s) [2s + \nu_w([0, s]) - \frac{1}{4} \int \nu_w(du) \nu_w(d(u+s))] ds - (w^2/2) R(0).$$

By scaling, the bracket equals $w h(s/w)$, which is at least $8w/3$ for every $s > 0$ by Lemma 4.6. Since $R \geq 0$ and $\int_0^\infty R = 1/2$, the bound is at least $(8w/3) \cdot 1/2 - (w^2/2)a$. ■

Theorem 4.1 follows by the continuum reduction of Part III, and the hypothesis $b > 0$ of the smoothing inequality is automatic, since $b \geq 8/(9a) > 0$.

Remarks. Three features of the argument stand out. First, the multiplier depends on the certificate only through a , just as the atom $2/a$ of the one-atom bound did. This is how the proof escapes the obstruction noted in the 25 September manuscript: for a fixed multiplier the bound on $a \cdot b$ is a product of two positive affine functions of R , hence minimized at a single kernel, and no fixed multiplier certifies $8/9$ on every kernel. Second, symmetry is never used, and mixtures are handled through R with no reduction to single kernels. Third, when $a \leq 4/3$ the width $w = 4/(3a)$ is at least 1, so the kernels see only the first window of μ_w , where the weight is constant; (4.2) with μ_w then equals $4w/3 - (w^2/2)a$ exactly, for every kernel. The renewal multiplier cannot distinguish kernels of equal energy.

Independent check. Solving the renewal equation on a grid of step 10^{-4} reproduces $\int (r - 2)^2 = 1/3$ to 2×10^{-5} and $\int (r - 2) = -1/3$ to 5×10^{-5} , the formula of Lemma 4.5 to 4×10^{-5} on $[0, 1]$, and $h(s) = 8/3$ on $(0, 1)$ to 5×10^{-5} ; on $[1, 12]$ the smallest value of h is $11/3$, at $s = 1+$ (Appendix A.3).

Part V · Sharpness: the extremal certificate and a new constant

The lower bound of Part IV is attained. The certificate that attains it was found by solving (3.6) with the numerically observed multiplier μ_1 . By the renewal identity, (3.6) for $u \geq 1$ follows from its restriction to $[0,1]$, where it becomes a linear integral equation for R whose solution is

$$R^*(s) = (2/3)(1 - |s|)^2(2 + |s|), \quad |s| \leq 1, \quad (5.1)$$

the autocorrelation of the single linear density $\rho(x) = 2x$ on $[0,1]$. That density is asymmetric, which is why symmetric certificates approach it only through mixtures.

The extremal continuum certificate

Theorem 5.1. Let $\rho(x) = 2x$ on $[0,1]$, and let r be the renewal density of $U[0,1]$. At the end whose covering condition involves the reflected kernel $2(1 - x)$, take $\omega' \equiv 1$. At the other end take $\omega = \frac{1}{4}(\mu_1 * \rho)$, that is

$$\omega(t) = e^{-t} - 1 \quad (0 \leq t < 1), \quad \omega(t) = \int_0^1 x r(t - x) dx \quad (t \geq 1). \quad (5.2)$$

This is a continuum certificate with $a = 4/3$ and $b = 2/3$, so $a \cdot b = 8/9$ and $\gamma = 2\sqrt{2}/3$.

Proof. A direct computation on $0 \leq u \leq 1$, using $r(v) = e^{-v}$ on $[0,1]$ and $r(v) = e^{-v} - v e^{-(v-1)}$ on $[1,2]$, gives $(\mu_1 * R^*)(u) = 4$. The renewal identity $\mu_1 = 2\delta_0 + U * \mu_1$ then gives $(\mu_1 * R^*)(u) = \int_{-} \{u-1\}^+ \mu_1 * R^*(u)$ for $u \geq 1$, since R^* vanishes there, so $\mu_1 * R^* \equiv 4$ on $[0, \infty)$. As $\int \rho(x)\omega(u+x) dx = \frac{1}{4}(\mu_1 * R^*)(u)$, the covering condition holds with equality everywhere at that end. At the other end $\frac{1}{4}(\mu_1 * \bar{\rho}) \equiv 1$ — for $0 \leq t < 1$ it is $(1 - t) + \int_0^t e^{-v} (1 - t + v) dv = 1$ — so $\omega' \equiv 1$ is both feasible and of the form required in Lemma 4.2, with equality in its covering condition. The equality conditions of Lemma 4.2 therefore hold with $\mu = \mu_1$, and since R^* lives inside the first window, the dual value is exactly $4/3 - \frac{1}{2}(4/3) = 2/3$, as in Theorem 4.7 with $w = 1 = 4/(3a)$. Finally $\omega - 1$ decays exponentially by Lemma 4.4(b). ■

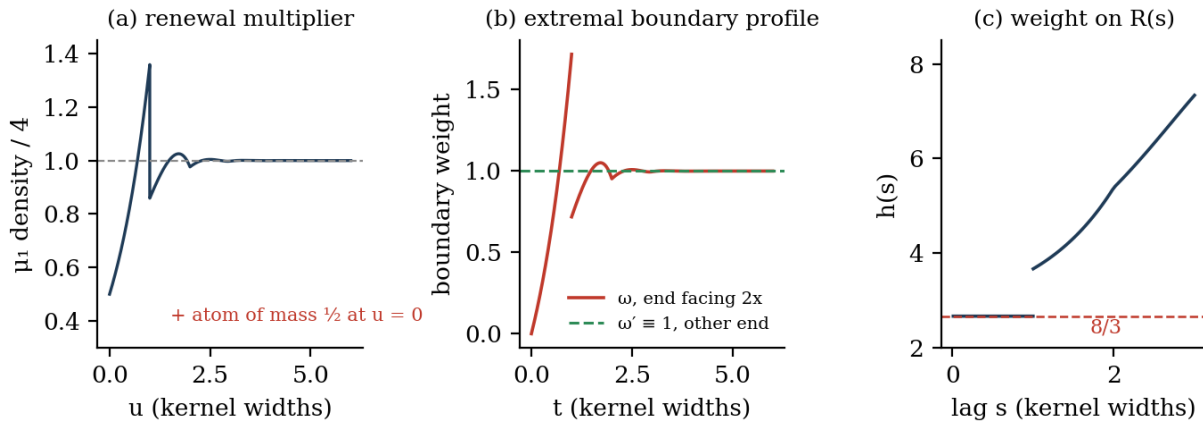


Figure 4. (a) The renewal multiplier μ_1 of (3.7), density divided by the flat value 4, plus an atom of mass $\frac{1}{2}$ at the origin in the same units; the optimal multipliers of the numerically optimal certificates converge to it at rate $1/m$. (b) The boundary profiles of the extremal certificate of Theorem 5.1: $e^{-t} - 1$ across the first kernel width at the end facing $2x$, a jump to $e - 2$ at $t = 1$, and decaying oscillations; $\omega' \equiv 1$ at the other end. (c) The weight $h(s)$ of Lemma 4.6 that the multiplier places on the autocorrelation: exactly $8/3$ on the first window, $11/3$ and rising beyond it.

The boundary profile discounts the outermost cells completely, as the optimized symmetric certificates did approximately. Kernel, profile and multiplier are tied together — $2x$ is the density of the maximum of two independent uniform variables — but that is an observation, not an explanation. **Independent check:**

numerically $\int_0^\infty (\omega^2 - 1) = -0.33351$ against $-1/3$, and the covering integral equals 1 to within 10^{-4} at every shift tested (Appendix A.3).

Discretization

Proposition 5.2. For every $m \geq 2$ there is a two-sided certificate with one kernel at resolution m and boundary depth $L = \max(2, \lceil (2/\kappa) \log m \rceil)$, with $a = 4/3 - 1/(3m^2)$ and $a \cdot b \leq 8/9 + C(\log m)/m^2$, where C depends only on the constants C_0, κ of Lemma 4.4(b). Hence $\gamma^* = 2\sqrt{2}/3$.

The proof discretizes (5.2) by cell averages and repairs the covering deficits with a uniform increment $\eta = 4/m^2 + \sup_{\{t \geq L-1\}} |\omega(t) - 1|$ on the boundary vector. At interior shifts the deficit is at most $4/m^2$, since the kernel and the profile deviate from their cell averages by at most $1/m$ and $4/m$. The delicate shifts are the last ones, where only a few kernel cells fall inside the box — at the very last, only the first, of mass $1/m^2$. There the deficit carries the same small in-box mass as the effect of the increment, because every in-box cell sits at depth at least $L - 1$, where the profile is exponentially close to 1; so an increment of order m^{-2} suffices. The factor $\log m$ comes only from spreading η over the whole box; a repair localized to the interior deficits would give $O(m^{-2})$, matching the exactly optimal certificates below. Numerically the construction is feasible at $m = 32, 64$ and 128 , with interior deficits at most 1.8×10^{-5} at $m = 128$ against the bound $4/m^2 = 2.4 \times 10^{-4}$.

Exactly optimal discretizations and the certified constant

With the exactly optimal boundary vectors in place of cell averages, the discrete values behave as cleanly as one could hope. The kernel $p_i = (2i + 1)/m^2$ has $a = 4/3 - 1/(3m^2)$, and with $L = 8$ the boundary program gives $b = 2/3 + 1/(3m^2)$ to within 3×10^{-8} for $m \leq 512$, so that $a \cdot b = 8/9 + 2/(9m^2) - 1/(9m^4)$ to that accuracy. The reversed end costs nothing: its optimal boundary vector is identically 1.

m	γ	$b - 2/3$	$1/(3m^2)$
32	0.9429240903	3.2555×10^{-4}	3.2552×10^{-4}
64	0.9428378328	8.1413×10^{-5}	8.1380×10^{-5}
128	0.9428162572	2.0377×10^{-5}	2.0345×10^{-5}
256	0.9428108625	5.1183×10^{-6}	5.0863×10^{-6}
512	0.9428095137	1.3035×10^{-6}	1.2716×10^{-6}

Table 8. The discretized linear kernel at $L = 8$ with exactly optimal boundary vectors at both ends. The $m = 32, 64$ and 128 rows were reproduced independently for this report with a separate quadratic-program solver, to 10^{-10} in γ (Appendix A.3). The certificates at $m = 256$ and 512 , rationalized and verified in exact arithmetic on 4098 and 8194 covering inequalities, give $\gamma < 0.942810871$ and $\gamma < 0.942809522$.

Theorem 5.3 (computational). The verified two-sided certificate with the linear kernel at $m = 512, L = 8$ has $a = 1.3333320618\dots$, $b = 0.6666679802\dots$ and $\gamma = 0.9428095208\dots < 0.942809522$. Consequently, by Proposition 3.1,

$$h(N) \leq N^{1/2} + 0.9428096N^{1/4} + O(1), \quad (5.3)$$

with the $O(1)$ term asymptotically $ab/2 \approx 0.444$ by (3.4). This is the smallest constant certified so far — 6.8×10^{-4} below Hou and Zhao's 0.9434926 — and it lies 4.8×10^{-7} above the proved limit, so no certificate in the framework can improve it by more than that.

Two structural facts

b depends only on the autocorrelation. For two-sided certificates the optimal b depends on the kernels and mixing weights only through $R = \sum_k \lambda_k p_k \star \tilde{p}_k$, as does $a = R(0)$. At a single end the boundary cost also

depends on the kernels' means, and these contributions cancel between the two ends. The two-sided problem is therefore to minimize $R(0) \cdot b(R)$ over the closed convex hull of autocorrelations of probability densities on $[0,1]$, with b a convex function of R , being a supremum of affine functionals. In this language Lemma 4.3 says $R(0) \cdot \int_0^1 |s| R(s) ds \geq \pi^2/32$, Theorem 4.7 says $R(0) \cdot b(R) \geq 8/9$, and Theorem 5.1 says the minimum $8/9$ is attained at R^* .

The symmetric framework. Symmetric certificates cannot contain the extremal kernel. Theorem 4.1 bounds them below by $2\sqrt{2}/3$, and the best found gives $\gamma < 0.942810893$ (verified; $m = 512$, $L = 4$). Their $L = 4$ scan extrapolates to the same 0.9428091 as the two-sided scan, and deepening the boundary from $L = 4$ to $L = 8$ at $m = 32$ improves the symmetric optimum by only 6×10^{-8} . Equality $\gamma^*_{\text{sym}} = 2\sqrt{2}/3$ would follow if R^* lay in the closed convex hull of autocorrelations of symmetric probability densities on $[0,1]$. Whether it does — and hence whether the original Hou–Zhao framework reaches the limit — is open.

Consequence for Erdős Problem 30

bound	γ	year
Erdős–Turán	1	1941
Lindström	1	1969
Balogh–Füredi–Roy	≈ 0.998	2021
O'Bryant	0.99703	2022
Carter–Hunter–O'Bryant	0.98183	2023
Carter, Georgiev, Gómez-Serrano, Hunter, O'Bryant, Tao, Wagner (announced)	0.97633	2026
Hou–Zhao, arXiv v1	0.94601	2026
Hou–Zhao, arXiv v3 (8 kernels)	0.94349	2026
independent 11-kernel certificate	0.94301	2026
this work, Version 3 (symmetric, $m = 512$)	0.94282	2026
this work, Version 4 (linear kernel, $m = 512$, $L = 8$)	0.9428096	2026
limit of the method (Theorem 4.1, Proposition 5.2)	$2\sqrt{2}/3 = 0.9428090\dots$	—

Table 9. The second-order constant. Years are those of first public appearance. Hou and Zhao's version 3 certifies $\gamma_0 = 0.9434925907\dots$; their version 1 (July 2026) gave 0.94601. The entire remaining room in the method below the Version 4 certificate is 4.8×10^{-7} .

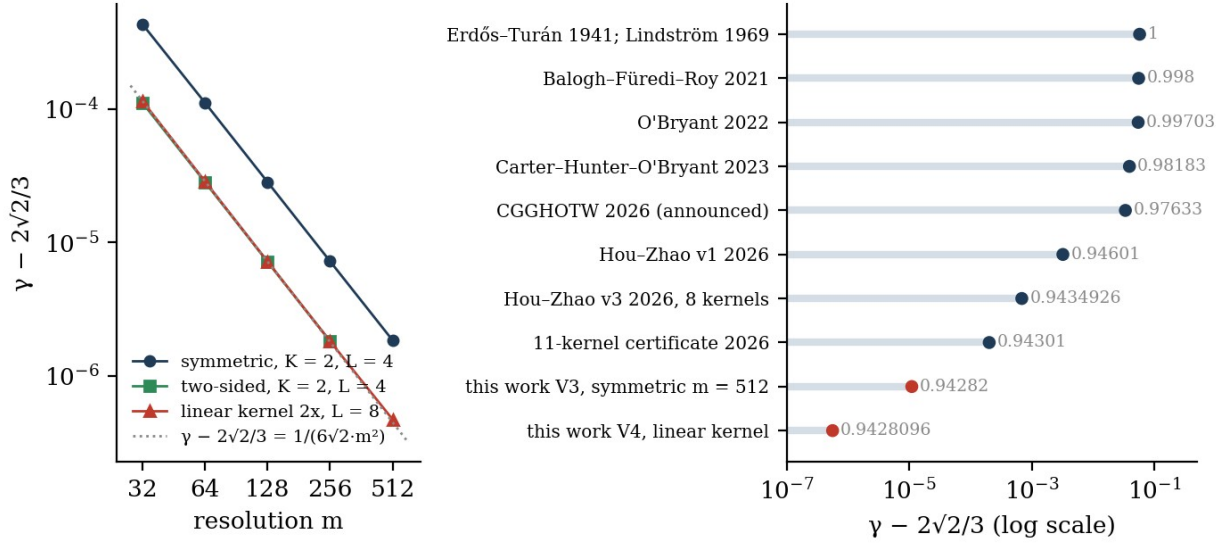


Figure 3. Left: distance above $2\sqrt{2}/3$ of the verified values of the optimized certificates and the exact quadratic-program values of the linear kernel, against the resolution m , with the reference line $1/(6\sqrt{2}m^2)$ that corresponds to $ab - 8/9 = 2/(9m^2)$. Right: the second-order constant through history, as distance above the proved limit on a logarithmic scale.

What an improvement would have to retain. The framework reaches its limit because the argument discards two quantities: the variance term in Cilleruelo's identity for the number of representations, which is nonnegative and thrown away when the energy is bounded, and the slack in the bound $d(h) \leq 1$ on the multiplicity of small differences. Any argument that goes below $2\sqrt{2}/3$ must retain at least one of them; Part VI shows that retaining both at the level of pair statistics is not enough. The qualitative conclusion, that arguments of this shape cannot reach $o(N^{1/4})$ let alone N^ϵ , already followed from any positive lower bound, including the $\pi/(4\sqrt{2})$ of the flat multiplier; what Version 4 adds is the exact location of the wall.

Part VI · Keeping the variance term: the pair-level relaxation

The natural question after Part V is whether the method has an extension that keeps what it discards, and what that extension's limit is. This part answers it at the first natural level: the extension is a semidefinite relaxation of the Sidon condition, and its second-order constant is again $2\sqrt{2}/3$. Throughout, $A \subset \{0, \dots, N-1\}$ is a Sidon set with k elements, a two-sided certificate at resolution (m, L, K) is fixed, and $H = mh$ with $N \geq 2LH$; K_r are the smoothing kernels, $S_r = \mathbb{1}_A * K_r$ the smoothed set, Q_r the weight functions, C_r the autocorrelation of K_r , and $c(x)$ the covering sum at x , as in the proof of the smoothing inequality.

The identity behind the smoothing inequality

Proposition 6.1. Put $c(A) = \sum_{\{x \in A\}} c(x)$, $\Delta(A) = \sum_r \lambda_r \sum_{\{d \neq 0, d \notin A-A\}} C_r(d)$ and $V(A) = (\sum_r \lambda_r \|Q_r\|^2) (\sum_r \lambda_r \|S_r\|^2) - c(A)^2$. Then $c(A) \geq k$, $V(A) \geq 0$, $\Delta(A) \geq 0$, and

$$c(A)^2 + V(A) + (N-1+bH) \cdot \Delta(A) = (N-1+bH) \cdot (1 + (k-1)a/H). \quad (6.1)$$

The proof of the smoothing inequality discards exactly three quantities: the covering slack $c(A)^2 - k^2$, the Cauchy-Schwarz defect $V(A)$, and the difference slack $(N-1+bH)\Delta(A)$. For the flat kernel with $u = v \equiv 1$ ($a = b = 1$), $V(A)/(N+H-1)$ is the variance of the window counts of A divided by H^2 , and $H^2 \Delta(A) = \sum_{\{0 < |d| < H, d \notin A-A\}} (H - |d|)$ is the slack in $d(h) \leq 1$; in that case (6.1) is, up to normalization, Cilleruelo's identity.

The two discarded quantities are coupled: for fixed k and $c(A)$, (6.1) fixes $V(A) + (N - 1 + bH)\Delta(A)$, so a set that uses fewer small differences has a larger variance and conversely, and a lower bound for either alone is equivalent to an upper bound on k . Normalized as $\varphi(A) = V(A)H/((N - 1 + bH)ka)$, which is about 1 for Poisson-like fluctuations, (6.1) gives $\varphi(A) \leq 1 + t(bt - 2\gamma)/a + O(N^{-1/4})$ whenever $k \geq \sqrt{N} + \gamma N^{1/4}$ and $H = tN^{3/4}$. At the certificate's own scale $t = \sqrt{a/b}$ the right-hand side is $2(1 - \gamma/\sqrt{ab}) + O(N^{-1/4})$, so every Sidon set more than halfway to a certificate's bound is sub-Poissonian, $\varphi < 1$, at scale $\sqrt{a/b} \cdot N^{3/4}$. Measured with a discretized linear-kernel certificate, the optimal Golomb rulers with 12 to 20 marks have φ between 0.23 and 0.45, and most of them use every difference below H ; the full-period sets of Bose and Ruzsa have φ between 1.1 and 1.5; a uniformly random subset of the same size and length gives 1.57.

The pair-level relaxation

Definition 6.2. For $N \geq 2$ let $\hat{h}_2(N)$ be the value of the semidefinite program that maximizes $\sum_x f(x)$ over $f \in \mathbb{R}^N$ and symmetric $\Gamma \in \mathbb{R}^{N \times N}$ subject to

$$\begin{aligned} & \text{(i) } [[1, f^T], [f, \Gamma]] \geq 0; \quad \text{(ii) } \Gamma(x, x) = f(x); \quad \text{(iii) } 0 \leq f \leq 1, \Gamma \geq 0 \text{ entrywise;} \\ & \text{(iv) } \sum_x \Gamma(x, x+d) \leq 1 \text{ for } 1 \leq d \leq N-1, \text{ with indices in } \{0, \dots, N-1\}. \end{aligned} \quad (6.2)$$

For a Sidon set A the pair $f = \mathbb{1}_A$, $\Gamma = \mathbb{1}_A \mathbb{1}_A^T$ is feasible with value k , so $h(N) \leq \hat{h}_2(N)$. Conditions (i)–(iii) are the level-one moment conditions of Lasserre and the pair conditions of Sherali and Adams for 0/1 variables; (iv) is the Sidon condition in expectation. The covariance $\Gamma - ff^T$ is kept positive semidefinite rather than discarded and each difference budget is kept separately, so both quantities of Proposition 6.1 become linear functionals of (f, Γ) , and the proof of the smoothing inequality becomes a dual solution.

Proposition 6.3. For every two-sided certificate and every $H = mh$ with $N \geq 2LH$, $\hat{h}_2(N)^2 \leq (N - 1 + bH) \cdot (1 + (\hat{h}_2(N) - 1)a/H)$. Consequently (3.4) holds with $h(N)$ replaced by $\hat{h}_2(N)$, and $\limsup_{N \rightarrow \infty} (\hat{h}_2(N) - \sqrt{N})/N^{1/4} \leq 2\sqrt{2}/3$.

Proof sketch. For feasible (f, Γ) with $\kappa = \sum f$, summing the covering condition against $f \geq 0$ and applying Cauchy–Schwarz in the weighted direct sum gives $\kappa^2 \leq (\sum_r \lambda_r \|Q_r\|^2)(\sum_r \lambda_r f^T \mathcal{C}_r f)$, where $\mathcal{C}_r(x, y) = C_r(x - y)$ is a Gram matrix. By (i), $f^T \mathcal{C}_r f \leq \text{tr}(\mathcal{C}_r \Gamma) = C_r(0)\kappa + \sum_{d \neq 0} C_r(d) \sum_x \Gamma(x, x+d) \leq C_r(0)\kappa + 1 - C_r(0)$, using (ii), $C_r \geq 0$ and (iv). The certificates of Proposition 5.2 give the limsup. ■ Only (i), (ii), (iv) and $f \geq 0$ are used.

The relaxation reaches $2\sqrt{2}/3$

The matching lower bound comes from explicit feasible points whose shape is read off from the equality conditions of Lemma 4.2. Near each end, a set that makes the Cauchy–Schwarz step tight must have density following $\frac{1}{4}\mu_1 = \frac{1}{2}\delta_0 + \frac{1}{2}r$: an atom, then a density that starts at half the bulk value, rises to $e/2$ at one kernel width and settles. The key fact is that this profile is invisible to short-range pair counts.

Lemma 6.4. Let $\sigma = \frac{1}{2}\delta_0 + \frac{1}{2}r\text{Leb}$ on $[0, \infty)$ and $\tau = \sigma - \text{Leb}$, a finite signed measure with $\tau([0, \infty)) = 1/3$. For $s > 0$ put $\mathcal{B}(s) = 2\tau([0, \infty)) - \tau([0, s]) + \int \tau(du)\tau(d(u+s))$. Then $\mathcal{B}(s) = s/2 + (8/3 - h(s))/4$, with h as in (4.5); in particular $\mathcal{B}(s) = s/2$ for $0 < s < 1$ and $\mathcal{B}(s) \leq s/2$ for $s \geq 1$.

Take a density on $[0, \sigma]$, lengths in kernel widths, that equals e at both ends. Its autocorrelation at a lag $s \in (0, 1)$ is $\mathcal{L} - s$ from the bulk plus $2\mathcal{B}(s) = s$ from the two ends, up to exponentially small terms — exactly $\mathcal{L}$. The ends supply exactly the pairs that the finite length removes; this is a second reading of the weight identity, Lemma 4.6.

The construction. Fix a bulk density δ , a scale $H = tN^{3/4}$, an atom fraction $\beta \in (0, 1]$ and $\ell = \lfloor 1/\delta \rfloor$. The profile ψ follows $\delta \cdot \frac{1}{2}r$ at each end, started at the centre of the atom, with bulk value δ ; the atom $\frac{1}{2}\delta_0$ is spread uniformly over the first and last ℓ sites, with mass $\beta\delta H/2$ at each end; and $f = \psi + \text{atoms}$. The covariance is $\Gamma = ff^T + D(I - \delta\Phi_\ell)D + \text{diag}(\varepsilon)$, where Φ_ℓ is the Toeplitz matrix with entries $(1 - |x - y|/\ell)_+$, $D = \text{diag}(\tilde{\psi}/\sqrt{\delta})$ with

$\tilde{\psi}$ a slightly reduced ψ , and $\varepsilon \geq 0$ restores the diagonal. Since the Fejér symbol of Φ_ℓ takes values in $[0, \ell] \subset [0, 1/\delta]$, the matrix $I - \delta\Phi_\ell$ is positive semidefinite and (i)–(iii) hold. The term $D(I - \delta\Phi_\ell)D$ is a covariance with a triangular hole: each bulk point gives up about one partner at lags below ℓ , which makes the bulk hyperuniform in the sense of Torquato and Stillinger.

Why the scale is $\sqrt{2}N^{3/4}$. Each atom contributes about $(\frac{1}{2}\delta H)^2$ ordered pairs at lags below ℓ , with the same triangular profile as the hole, so (iv) at a lag $d < \ell$ reads, to leading order, $\delta^2 N + (1 - d/\ell) \cdot [2(\frac{1}{2}\delta H)^2/\ell - \delta^2 N] \leq 1$. The atoms refill the vacated lags exactly when $2(\frac{1}{2}\delta H)^2 = \ell \approx \sqrt{N}$, that is $H = \sqrt{2}N^{3/4}$ — the certificate’s own optimal scale $\sqrt{(a/b) \cdot N^{3/4}}$ for $a = 4/3$, $b = 2/3$, and no other. At that scale $\delta^2 N \approx 1$, and each end carries an excess $b/2 = 1/3$ in units of δH ($\frac{1}{2}$ from the atom and $-1/6$ from the continuous part), so $\Sigma f = \delta N + (2/3)\delta H + O(1) = \sqrt{N} + (2\sqrt{2/3})N^{1/4} + O(1)$.

N	t	β	$\delta\sqrt{N}$	$\Sigma_x f(x)$	$(\Sigma f - \sqrt{N})/N^{1/4}$	$\sqrt{N} + (2\sqrt{2/3})N^{1/4} - \Sigma f$
10^4	1.344	0.9936	1.00514	108.441	0.844	0.99
10^5	1.414	0.9698	1.00221	331.969	0.885	1.02
10^6	1.414	0.9837	1.00069	1028.794	0.911	1.02
10^7	1.414	0.9912	1.00021	3214.284	0.925	1.01

Table 10. Explicit feasible points of (6.2). The shortfall from $\sqrt{N} + (2\sqrt{2/3})N^{1/4}$ in the last column is constant to within 0.04 over three decades; the normalized excess in the sixth column approaches $2\sqrt{2/3} = 0.943$ slowly because the shortfall is $O(1)$. At $N = 10^7$, β was extrapolated from the smaller cases rather than optimized.

Theorem 6.5 (computational). For $N \in \{10^4, 10^5, 10^6, 10^7\}$ these points are feasible for (6.2), and $\hat{h}_2(N) \geq \sqrt{N} + (2\sqrt{2/3})N^{1/4} - 1.03$. Conditions (i)–(iii) hold by construction, with the hypotheses checked on the computed vectors. Condition (iv) was evaluated in double precision at every d by fast Fourier transform; the largest difference sum is at most $1 - 1.3 \times 10^{-8}$, and lowering δ by a relative 10^{-7} leaves every sum at most $1 - 2 \times 10^{-7}$ at a cost of at most 3.3×10^{-4} in Σf . An explicitly formed feasible point at $N = 3000$ has a moment matrix with smallest eigenvalue 6.1×10^{-5} , and direct summation of (iv) agrees with the transform to 4×10^{-16} . This is a double-precision computation with an explicit safety margin, not an exact-arithmetic verification.

Conjecture 6.6. $\hat{h}_2(N) = \sqrt{N} + (2\sqrt{2/3})N^{1/4} + O(1)$. The upper half is Proposition 6.3. For the lower half the construction satisfies (iv) up to four errors — the finite width of the atoms, the reduction $\tilde{\psi} < \psi$ where the profile exceeds the bulk density, atom–bulk pairs at short lags, and discretization — each of relative size $O(N^{-1/4})$ in the budget of a single lag. They force $\beta = 1 - O(N^{-1/4})$, and $1 - \beta \approx 0.030, 0.016$ and 0.009 is observed at $N = 10^5, 10^6$ and 10^7 ; they cost $O(1)$ in Σf . Writing out this error analysis would prove the conjecture; it has not been done.

Structure of the optimum at moderate N

The program (6.2) was solved for $20 \leq N \leq 180$ with Clarabel through CVXPY, after block-diagonalizing by the reflection $x \mapsto N - 1 - x$. Table 11 compares it with $h(N)$ and with the finite- N smoothing bound $s(N)$ given by the linear kernel with exactly optimal weights, minimized over H .

N	h(N)	$\hat{h}_2(N)$	s(N)	s(N) – $\hat{h}_2(N)$
20	6	6.4291	6.5822	0.1531
32	7	7.9189	8.0726	0.1537
40	8	8.7405	8.8939	0.1534
50	9	9.6469	9.7993	0.1524
64	10	10.7568	10.9089	0.1521
80	11	11.8671	12.0198	0.1527
100	12	13.0938	13.2489	0.1551
120	13	14.1959	14.3522	0.1563
140	14	15.2037	15.3612	0.1575
160	15	16.1372	16.2958	0.1586
180	16	17.0104	17.1699	0.1595

Table 11. The pair-level relaxation against the smoothing bound. The gap stays between 0.152 and 0.160 while N grows ninefold; a difference of 0.05 in the second-order constant would open it by about 0.08 over this range. For this report $\hat{h}_2$ was re-solved at $N = 12, 20, 32$ and 40 without the symmetry reduction, reproducing all printed values to four decimals (Appendix A.3).

Two warnings apply. Second-order constants cannot be fitted reliably from $N \leq 180$: fitting $\sqrt{N} + \gamma N^{1/4} + c_0 + c_1 N^{-1/4}$ to $s(N)$ itself, whose limit is 0.943, returns anything from 0.85 to 0.96, so the asymptotic claim rests on Theorem 6.5, not on these fits. And Clarabel flagged some instances as possibly inaccurate, although the returned solutions satisfy every constraint of (6.2) to within 10^{-9} .

The optimal dual is the smoothing certificate. The optimal multipliers y_d of the difference constraints (iv) fit $y_d \approx cR^*(d/H_{\text{fit}})$, with R^* the autocorrelation (5.1) of $2x$. The relative L^2 misfit is 6.1%, 4.1%, 3.3%, 2.9%, 2.6% and 2.5% at $N = 32, 50, 64, 80, 100$ and 120 , and the fitted scale $H_{\text{fit}} = 22.2, 30.7, 35.3, 39.6, 47.7, 55.6$ tracks the optimal H of $s(N)$: 22, 30, 35, 41, 49, 56. The only visible departure is at $d = 1$. Lindström’s flat kernel fits more than twice as badly (5.8% at $N = 120$).

The optimal primal is the configuration above. At $N = 120$ the endpoints have $f = 0.98$, and the first six sites at each end carry total mass 2.49 against $N^{1/4}/\sqrt{2} = 2.34$ predicted, followed by a depleted stretch. In the middle half $\Gamma(x, x+d) = 0$ for $1 \leq d \leq 6$, against a predicted radius $\sqrt{N}/2 \approx 5.5$, and $\Gamma \approx ff^T$ from $d = 8$ on; the covariance hole sums to about -1 per point; and (iv) is tight for every $d \leq 56$, against $\sqrt{2}N^{3/4} \approx 51$. Adding the distinct-sums constraints and the Sherali–Adams inequalities $\Gamma(x, y) \leq f(x)$ and $\Gamma(x, y) \geq f(x) + f(y) - 1$ lowers the optimum by only 0.019 at $N = 12$ and 0.006 at $N = 100$.

Beyond pairs

Lemma 6.7. In any relaxation with triple variables $T(x, y, z)$ that contains the lifted budgets $\sum_x T(x, x+d, z) \leq f(z)$ and $\sum_x [\Gamma(x, x+d) - T(x, x+d, z)] \leq 1 - f(z)$, the equality $\sum_x \Gamma(x, x+d) = 1$ forces $\sum_x T(x, x+d, z) = f(z)$ for every z . Wherever a difference budget is full, the number of pairs at that difference must be uncorrelated with every point indicator. In the extremal configuration every budget below H is full, and the independent atoms of the construction violate this.

Deterministic atoms. Deterministic atoms have constant indicators and satisfy every constraint involving them at every level. Take for the two atoms the two halves, m points each, of the Ruzsa set for the smallest prime $p \geq 2m + 1$, whose difference sets are disjoint, and modify the construction: replace the hole matrix by the Toeplitz matrix with entries 1 at lags in $\{0\} \cup D$, where D is the atoms’ difference set; cap the profile at $c = 1/(1 +$

$2|D|$); and deplete a zone of length $\text{span} + \max D$ at each end. The result is a family of feasible points whose atoms are genuine Sidon sets (Table 12), checked in exact arithmetic at every lag that carries an atom pair.

N	m	t	$\delta\sqrt{N}$	$\Sigma_x f(x)$	$\sqrt{N} + (2\sqrt{2/3})N^{(1/4)} - \Sigma f$
10^4	7	1.2	1.0356	107.324	2.10
10^5	13	1.25	1.0358	330.962	2.03
10^6	22	1.414	1.0060	1027.411	2.40
10^7	40	1.5	1.0041	3213.196	2.10

Table 12. Feasible points of (6.2) whose atoms are genuine Sidon sets. The shortfall is $O(1)$, about 1.1 larger than in Table 10, and fluctuates with the integer choice of m . Any obstruction beyond the pair level must therefore live in the bulk, which must use every difference below H once in expectation, except the roughly $\sqrt{N}/2$ differences left to the atoms.

Three points on the interval. The three-point relaxation $\hat{h}_3(N)$ adds triple variables with positive semidefiniteness of the moment matrices localized at $z \in A$ and at $z \notin A$ for every z , the lifted budgets, $T(x, x+d, x+2d) = 0$ (a Sidon set contains no three-term progression), the Sherali–Adams bounds, and $\Gamma \leq f$.

N	12	18	20	26	32	40
$h(N)$	5	6	6	7	7	8
$\hat{h}_2(N)$	5.1194	6.1340	6.4291	7.2237	7.9189	8.7405
$\hat{h}_3(N)$	5.0000	6.0149	6.3271	7.1370	7.8554	8.6916
$\hat{h}_2 - \hat{h}_3$	0.119	0.119	0.102	0.087	0.064	0.049

Table 13. Three-point information lowers the relaxation by an amount that shrinks with N , whereas a second-order improvement would require growth like $N^{(1/4)}$. Solved with Clarabel, and with SCS for $N \geq 32$; the two agree to four decimals at $N = 26$. At $N = 20$ the lifted budgets and the localized positive semidefiniteness do most of the work, synergistically: each alone recovers about a third of the reduction.

A cyclic model of the bulk. To isolate the bulk, consider translation-invariant relaxations on $\mathbb{Z}_n$ with $n = q^2 - 1$, with the Sidon condition taken in $\mathbb{Z}_n$ and every difference in a symmetric forbidden set F of $q - 2$ nonzero residues excluded, so that the counting bound is exactly q and the pair-level relaxation attains it. Three forbidden sets of the same size are compared: the subgroup of nonzero multiples of $q + 1$, which are exactly the residues Bose’s sets miss; short lags; and random lags. The four-point relaxation uses the moment matrix indexed by subsets of size at most 2, with exact zeros on every set of at most four elements containing a repeated or forbidden difference; it is block-diagonalized by the Fourier transform on $\mathbb{Z}_n$ and returns exactly 4 and 5 on the Singer moduli 13 and 21.

$n = q^2 - 1$	forbidden set	pair	three-point	four-point	largest genuine set
24	subgroup	5	5	5.0000	5
24	short lags	5	5	4.2983	4
24	random lags	5	5	4.2983	4
48	subgroup	7	7	7.0000	7
48	short lags	7	7	6.9720	6
48	random lags	7	7	6.9720	6
63	subgroup	8	8	8.0000	8
63	short lags	8	8	8.0000	7
63	random lags	8	8	8.0000	7

Table 14. The cyclic model; the largest genuine sets were determined by exhaustive search. At $n = 63$ the four-point relaxation was solved with SCS, feasible to 3×10^{-8} ; since the counting bound caps every relaxation at q , the values 8 are exact to that order.

Only the subgroup admits a set of size q ; for the other two shapes the largest genuine set has $q - 1$ elements, yet the pair and three-point relaxations return q for every shape. The four-point relaxation, which contains the Sidon condition in its native form $a + b = c + d$, closes 70% of the gap between the counting bound and the truth at $n = 24$, 2.8% at $n = 48$ and none of it at $n = 63$, and where it cuts it separates the subgroup from the other two shapes, not short lags from random ones. The obstruction is arithmetic and global: invisible at the pair and three-point levels, and what the four-point level captures of it disappears by $n = 63$. Three small sizes cannot settle whether any fixed level of the moment hierarchy captures it as n grows, but they suggest not.

Consequences

The level-one moment relaxation is the extension of the smoothing method that keeps the variance term and the difference slack. Its dual is a semidefinite program of which smoothing certificates are feasible points — the kernel autocorrelations are the multipliers of (iv), and the Cauchy–Schwarz direction is the positive semidefinite multiplier of (i) — and at moderate N its optimal dual is the extremal smoothing certificate up to local corrections. Its second-order constant is $2\sqrt{2}/3$: at most that by Proposition 6.3, and within 1.03 of it up to $N = 10^7$ by Theorem 6.5. Keeping the variance term therefore gains only $O(1)$ as long as the argument sees one- and two-point statistics, and the evidence above suggests that three- and four-point statistics add little. In the cyclic model, what separates the relaxations from genuine Sidon sets is a design-type question — whether a near-perfect difference set with a prescribed set of missing differences exists — and an argument that goes below $2\sqrt{2}/3$ probably has to use arithmetic structure of that kind.

Part VII · Where this leaves the programme

The programme was laid out in five stages. Their status after this work:

stage	content	status
Stage 0	Construction survey across Singer, Bose, Ruzsa families and all characteristics	Complete. Outcome A: no power-law growth; bet the conjecture is true.
Stage 1	Prove the largest gap over multipliers is not $O(q)$, refuting $h(N) = \sqrt{N} + O(1)$	Open, and harder than expected. Character sums cannot certify even two-point independence of the zero events at the relevant scale; needs an idea about runs in m -sequences that is not a character sum.
Stage 2T(a)	Determine the floor of the smoothing framework	Closed. $\gamma^* = 2\sqrt{2}/3$: lower bound proved for every certificate (Part IV), attained by two-sided certificates (Part V); verified certificate 0.9428096. Symmetric floor in $[2\sqrt{2}/3, 0.942810893]$, equality open.
Stage 2T(b)	Extract the forced structure of a hypothetical dense Sidon set	Partial. In the literature (Cilleruelo; Ortega–Prendiville) at the $N^{1/4}$ level. New: sets more than halfway to a certificate’s bound are sub-Poissonian at scale $\sqrt{(a/b) \cdot N^{3/4}}$; the extremal pair-level configuration is two atoms plus a hyperuniform bulk (Part VI).
Stage 2T(c)	Find the second-order constraint separating $\log N$ from N^c	Open. New negative evidence: keeping the variance term at the pair level gains only $O(1)$; three-point gains shrink with N ; the four-point level captures a vanishing share of the cyclic-model obstruction (Part VI).

Table 15. Stage status.

Stage 2T(a) is closed. Version 3 named the route — a dual multiplier richer than a single atom, aimed at the target $ab \geq 8/9$ — and that is what was carried out: the multiplier is the renewal measure of the uniform distribution, rescaled with the certificate’s energy. The open questions, roughly in order of tractability:

- Is $\gamma^*_{\text{sym}} = 2\sqrt{2}/3$? Equivalently, does R^* lie in the closed convex hull of autocorrelations of symmetric probability densities on $[0,1]$? This decides whether the original Hou–Zhao framework reaches the limit.
- Conjecture 6.6: write out the $O(N^{(-1/4)})$ error analysis of the pair-level construction.
- A conceptual proof of the weight identity, Lemma 4.6, which is at present a calculation; Lemma 6.4 gives one reading of it.
- Do other smoothing schemes — windows of varying width, weighted energies, higher-order Cauchy–Schwarz — have similarly explicit limits?
- Stages 1 and 2T(c), as before: properties of runs in m -sequences beyond Weil-type character-sum bounds, and an argument that uses the arithmetic structure of dense Sidon sets rather than the statistics of a fixed number of their points.

Three things are worth avoiding: further certificate search inside the framework, where the proved room is below 5×10^{-7} ; further exact- $h(N)$ tabulation beyond $N = 586$, which costs years of distributed computation for one more point on a sawtooth the Singer data already explain; and higher levels of the moment hierarchy as a route below $2\sqrt{2}/3$ unless they are combined with arithmetic input, given Tables 13 and 14.

Status of the write-ups. Two manuscripts record Parts III–VI. “A barrier for vector-valued smoothing certificates for Sidon sets” (25 September 2026) proves $\gamma^* \geq 0.899124$, identifies the optimal multiplier and the linear-kernel certificate, and conjectures $\gamma^* = 2\sqrt{2}/3$. It is superseded by “Vector-valued smoothing for Sidon sets cannot beat $2\sqrt{2}/3$ ” (26 September 2026), which proves that conjecture for two-sided certificates, gives the full proof of the two-sided smoothing inequality, and adds the pair-level relaxation. Material from the earlier

manuscript that the later one does not carry — the full proof of the cosine-arch inequality, the table of multiplier truncations and the multiplier family fitted per certificate — is recorded in Parts III and IV here. Before submission the later manuscript still needs its author block and the deposit of code and certificates that it promises.

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Appendix A · Data, checks and file inventory

A.1 Verified certificates

certificate	m	L	K	inequalities	a	b	γ
symmetric, original scan	32	4	2	129	1.294133696	0.6874915809	0.94324229
symmetric, original scan	64	4	2	257	1.313278418	0.6770076151	0.94292072
symmetric, original scan	128	4	2	513	1.3216771522	0.6725867335	0.94283748
symmetric, original scan	256	4	2	1025	1.3262775741	0.6702236072	0.94281628
symmetric, original scan	512	4	2	2049	1.3264253589	0.6701415970	0.94281112
symmetric, re-optimized	512	4	2	2049	1.3286071522	0.6690407889	0.9428108915
two-sided, optimized	32	4	2	258	1.3131380851	0.6770798880	0.9429206688
two-sided, optimized	64	4	2	514	1.3230863497	0.6718702582	0.9428374024
two-sided, optimized	128	4	2	1026	1.3281707583	0.6692681652	0.9428162103
two-sided, optimized	256	4	2	2050	1.3305653396	0.6680561258	0.9428108643
linear kernel 2x	256	8	1	4098	1.3333282471	0.6666717951	0.9428108696
linear kernel 2x	512	8	1	8194	1.3333320618	0.6666679802	0.9428095208

Table A1. Every certificate was rationalized, its boundary program re-solved for the rational kernels, its boundary vectors rounded and repaired by a uniform increment, and all covering inequalities checked over $\mathcal{Q}$ ($Lm + 1$ for symmetric, $2(Lm + 1)$ for two-sided certificates); a , b and γ^2 were then evaluated exactly. Mixing weights λ_1 : 0.3592, 0.3440, 0.3414, 0.3397, 0.3396 for the original symmetric scan, 0.3383 for the re-optimized certificate, and 0.1475, 0.1488, 0.1488, 0.1492 for the two-sided scan. Because the optimum is flat in some directions, a and b of the optimized rows are not individually converged; only γ is. For the linear kernel, $a = 4/3 - 1/(3m^2)$ exactly.

A.2 Stage 0 survey (stage0_results.csv)

174 rows: 34 Singer, 29 Bose and 111 Ruzsa instances. Columns are family, q , p , e , n , k , maxgap, maxgap/ q , the extreme-value prediction and ratio, the excess at zero drops and at the best drop count, the best drop count d , the resulting interval length N , and the full excess profile for $d = 0 \dots 8$. The excess at zero drops is $k - \sqrt{(n - \text{maxgap} + 1)}$. Representative rows at the top of the range:

family	q	n	k	max gap	EV ratio	excess (cut)	excess (best arc)	d	N
singer	1009	1019091	1010	20503	1.097	10.71	11.50	5	987047
singer	1024	1049601	1025	22134	1.340	11.36	12.14	2	1021844
singer	1369	1875531	1370	29683	1.169	11.38	12.10	4	1833045
bose	1024	1048575	1024	20886	1.249	10.25	12.36	3	1017354
bose	1331	1771560	1331	24888	1.066	9.38	11.58	2	1735605
ruzsa	1447	2092362	1446	16574	0.898	5.24	6.90	1	2068136

Table A2. Selected rows from the top of the Stage 0 range, taken directly from stage0_results.csv; N is the interval length at the best drop count d . The Singer and Ruzsa rows printed in Version 3 did not match the file and are corrected here.

A.3 Independent checks for Version 4

what was checked	how	result
Renewal identities, Lemmas 4.4–4.6	renewal equation solved on a grid of step 10^{-4} over $[0, 40]$	$\int (r-2)^2 = 0.333312$, $\int (r-2) = -0.333383$; $A(t)$ formula to 4×10^{-5} on $[0,1]$; $h = 8/3$ on $(0,1)$ to 5×10^{-5} ; min h on $[1, 12] = 11/3$ at $s = 1+$
Extremal continuum certificate, Theorem 5.1	same grid	$\int (\omega^2 - 1) = -0.33351$; covering integral 1 to 10^{-4} for $u \in [0, 6)$; $\omega(1-) = e - 1$, $\omega(1+) = e - 2$
Linear-kernel boundary program at $L = 8$, Table 8	CVXPY with Clarabel, both ends solved separately	$\gamma = 0.942924090, 0.942837833, 0.942816257$ at $m = 32, 64, 128$; $b - 2/3$ as printed; reversed end optimal at $v \equiv 1$
Pair-level relaxation, Tables 11 and 13	CVXPY with Clarabel, no symmetry reduction	$\hat{h}_2 = 5.1194, 6.4291, 7.9189, 8.7405$ at $N = 12, 20, 32, 40$
Hou-Zhao constants	arXiv 2607.01169, current version and v1 abstract	$\gamma_0 = 0.9434925907\dots$, eight kernels, $m = 32$, $L = 4$; one symmetric kernel 0.9461473014; v1: 0.94601
Part I fits	optimal Golomb ruler lengths, $k \leq 28$	$\gamma = 0.6404$ (RMSE 0.3847); $c = 2.639$ (RMSE 0.4899); $\alpha = 0.2013$ on values, 0.1977 in log-log; $D(k)$ slope 1.426
Stage 0 summaries, Table 2 and Table A2	stage0_results.csv	Table 2 reproduced to ≤ 0.002 after excluding $q \leq 16$ (Singer) and $p \leq 43$ (Ruzsa), except the Bose odd-characteristic cut slope (0.765, printed 0.756 in V3)
Figure 5	quadrature with 2×10^5 points	cosine arch $0.308425 = \pi^2/32$; parabolic 0.308571; all other densities higher

Table A3. Checks carried out for this version, independently of the code that produced the original results; the scripts are in `v4_checks.py`.

A.4 Files

file	contents
Sidon_smoothing_limit.docx	manuscript, 26 September 2026: $\gamma^* = 2\sqrt{2}/3$, the two-sided smoothing inequality in full, the pair-level relaxation (Parts IV–VI)
Sidon_smoothing_barrier.docx	manuscript, 25 September 2026: $\gamma^* \geq 0.899124$, the optimal multiplier, the linear-kernel certificate, conjecture $\gamma^* = 2\sqrt{2}/3$; superseded
Erdos_Problem_30_Report_V3.docx, V3_changes.md	the previous version of this report and its change log
V4_changes.md	changes from Version 3 to Version 4
v4_checks.py	the independent checks of Table A3
v4_figures.py	regenerates Figures 1–5 from the values in this report and stage0_results.csv
FLOOR_RESULT.md	earlier standalone write-up of the numerical floor (superseded; its drop and ratio columns predate the Version 3 correction)
POSITIVE_INFIMUM.md	earlier write-up of the flat-multiplier bound (superseded)
floor_scan.csv, verified_m{32,...,512}_L4_R2.npz, cert_m512_*.txt	the original symmetric scan and its rationalized certificates
floor2.py, certify.py, verify_lemma.py, search.py, step.py, one.py, upsample.py, lscan.py	optimizer, exact-rational verifier and drivers for the symmetric scan
engine.py, run_all.py, run_rest.py, tail.py, ogr_table.py	Stage 0 engine and drivers, and the OGR comparison
stage0_results.csv, ogr_vs_algebraic_cuts.txt	the full Stage 0 survey (174 rows) and Table 1 as generated
two-sided optimizer, linear-kernel certificates, continuum evaluator, SDP and construction code	referenced in the manuscripts; to be deposited with the published version